

Conservative upwind finite-element method for a simplified Keller–Segel system modelling chemotaxis

NORIKAZU SAITO[†]

Faculty of Human Development, University of Toyama, 3190 Gofuku,
Toyama 930-8555, Japan

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Finite-element approximation for a non-linear parabolic–elliptic system is considered. The system describes the aggregation of slime moulds resulting from their chemotactic features and is called a simplified Keller–Segel system. Applying an upwind technique, first we present a finite-element scheme that satisfies both positivity and mass conservation properties. Consequently, if the triangulation is of acute type, our finite-element approximation preserves the L^1 norm, which is an important property of the original system. Then, under some assumptions on the regularity of a solution and on the triangulation, we establish error estimates in $L^p \times W^{1,\infty}$ with a suitable $p > d$, where d is the dimension of a spatial domain. Our scheme is well suited for practical computations. Some numerical examples that validate our theoretical results are also presented.

Keywords: finite-element method; parabolic–elliptic system; non-linear evolution equation; analytic semigroup; conservation of positivity; conservation of mass.

1. Introduction

We consider a non-linear parabolic–elliptic system for the functions $u = u(x, t)$ and $v = v(x, t)$ of $(x, t) \in \overline{\Omega} \times [0, J)$:

$$\begin{cases} \frac{\partial u}{\partial t} = \nabla \cdot (k \nabla u - \lambda u \nabla v) & \text{in } \Omega \times (0, J), \\ 0 = \Delta v - v + u & \text{in } \Omega \times (0, J), \\ k \frac{\partial u}{\partial \nu} - \lambda u \frac{\partial v}{\partial \nu} = 0, \quad \frac{\partial v}{\partial \nu} = 0 & \text{on } \partial \Omega \times (0, J), \\ u|_{t=0} = u_0 & \text{on } \Omega, \end{cases} \quad (1.1)$$

where $\Omega \subset \mathbb{R}^d$ ($d = 2, 3$) is a bounded domain with the boundary $\partial \Omega$, ν is the outer unit normal vector to $\partial \Omega$, $\partial/\partial \nu$ represents differentiation along ν on $\partial \Omega$, $u_0 = u_0(x) \geq 0$ is a given non-identically zero function defined on Ω and J, k, λ are positive constants.

The system (1.1) describes the aggregation of slime moulds resulting from their chemotactic features, where u denotes the density of the cellular slime moulds and v is the concentration of the chemical substance. Replacing the constants k, λ in the first equation of (1.1) with functions $k(u, v), \lambda(v)$ and taking

$$\lambda_0 \frac{\partial v}{\partial t} = \Delta v - \lambda_1(v)v + \lambda_2(v)u \quad (\lambda_0 > 0 : \text{a constant}, \lambda_1, \lambda_2 : \text{functions of } v)$$

instead of the second equation, we obtain the original system proposed by Keller & Segel (1970). **If diffusion of the chemical substance takes place much faster than that of the cellular slime moulds,**

[†]Email: saito@infsup.jp

the original Keller–Segel system can be reduced to (1.1). On the other hand, in some other contexts, we often obtain problems that possess quite similar structure to that of (1.1) including transient drift diffusion of electrons in semiconducting media (Biler *et al.*, 1994; Jüngel, 2001) and gravitational interaction of particles (Biler & Nadzieja, 1994). Therefore, the results of this paper are of use not only in mathematical biology but also in engineering and theoretical physics.

The mathematical study for (1.1) is well developed. In particular, Biler (1998) and Yagi (1997) showed unique solvability locally in time when $\partial\Omega$ and u_0 are sufficiently smooth. Apart from these fundamental results, many researchers elucidated the behaviour of solutions: relevant properties include aggregation, blow-up, chemotactical collapse and so on. Two survey articles, Horstmann (2003) and Horstmann (2004), review those mathematical results. A salient property of a solution (u, v) to (1.1) is the conservation of the L^1 norm:

$$\|u(t)\|_{L^1(\Omega)} = \|u_0\|_{L^1(\Omega)}, \quad t \in [0, J], \quad (1.2)$$

which is an immediate consequence of the conservation of positivity

$$u(x, t) > 0, \quad (x, t) \in \overline{\Omega} \times (0, J], \quad (1.3)$$

and the conservation of total mass

$$\int_{\Omega} u(x, t) dx = \int_{\Omega} u_0(x) dx, \quad t \in [0, J]. \quad (1.4)$$

Actually, the steady-state problem corresponding to (1.1) is reduced to a non-local eigenvalue problem by (1.2). Moreover, the value of $\|u_0\|_{L^1(\Omega)}$ is crucial to the global existence and blow-up of a solution. For example, in the case of $d = 2$, it is known that if $\|u_0\|_{L^1(\Omega)} < 4\pi\lambda^{-1}$, the solution exists globally in time, whereas if $\|u_0\|_{L^1(\Omega)} > 8\pi\lambda^{-1}$ and $\int_{\Omega} |x - x_0|^2 u_0(x) dx \ll 1$ with some $x_0 \in \Omega$, then the solution blows up in finite time. Furthermore, quantization of the blow-up mechanism can be observed. For further details, see Suzuki (2005).

The present paper has dual purpose. The first is to propose a finite-element scheme for (1.1) that satisfies the discrete analogues of (1.3) and (1.4). As stated above, the L^1 conservation is an essential requirement; it is desirable that approximate solutions preserve this property. In a continuous problem, derivations of (1.3) and (1.4) are simple. Indeed, (1.3) is a result of the maximum principle and (1.4) follows from

$$\frac{d}{dt} \int_{\Omega} u(x, t) dx = \int_{\Omega} \nabla \cdot (k \nabla u - \lambda u \nabla v) dx = \int_{\partial\Omega} \left(k \frac{\partial u}{\partial \nu} - \lambda u \frac{\partial v}{\partial \nu} \right) dS = 0.$$

In a discrete problem, however, some difficulties arise. First, as is well-known, we should choose our discretization parameters very carefully in order to guarantee the discrete version of (1.3). In a previous paper, Saito & Suzuki (2005), we discussed that issue and proposed finite-difference schemes that satisfy discrete versions of both (1.3) and (1.4). Those schemes made use of an upwind difference scheme and semi-implicit time discretization with step-size control. That is, at every discrete time step $t_n = \tau_1 + \cdots + \tau_n$, we adjust the time increment τ_n to obtain a positive solution. We intend to extend this strategy to finite-element methods. In doing so, however, we confront another issue. Upwind-type finite-element approximations usually destroy the conservation of total mass. To surmount this obstacle, Baba & Tabata (1981) proposed a conservative upwind finite-element approximation for a linear convection–diffusion equation. We combine our previous strategy with the upwind approximation of Baba & Tabata. Consequently, we realize (cf. Theorems 2.1 and 2.2) a finite-element approximation that satisfies discrete versions of both (1.3) and (1.4) for an arbitrary $h > 0$, the granularity parameter of

the spatial discretization, if the triangulation is of acute type. (Ω is assumed to be a convex polygonal domain.) Moreover, our scheme is well suited for practical computations.

The second purpose is to develop an error analysis. Let (u_h^n, v_h^n) be a solution of the finite-element scheme proposed below at a discrete time step t_n . In our scheme, the point-wise value of ∇v_h^{n-1} is used to determine the upwind nodal points at t_n . Therefore, we need to pay attention to the error $v_h^n - v(t_n)$ in the $W^{1,\infty}(\Omega)$ norm (uniformly in t_n). On the other hand, we have (cf. (6.13) and (6.14))

$$\|v(t_n) - v_h^n\|_{W^{1,\infty}(\Omega)} \leq Ch^{1-d/p} \|u(t_n)\|_{L^p(\Omega)} + C \|u(t_n) - u_h^n\|_{L^p(\Omega)}$$

for some $p \in (d, \mu)$. Here, $\mu > d$ is the shape constant of Ω ; it is defined through the condition (R) described below, which can be thought of as a quantification of the $W^{2,p}$ regularity property of a linear elliptic problem. In any case, it is crucial to study the error $u(t_n) - u_h^n$ in the L^p norm (uniformly in t_n). For these reasons, we shall develop the L^p theory for (1.1) and its finite-element approximation. In fact, if the triangulation is of acute type and is quasi-uniform, under some regularity hypotheses on u , we have (cf. Theorem 2.3)

$$\sup_{0 \leq t_n \leq J} \|u(t_n) - u_h^n\|_{L^p(\Omega)} \leq C(h^{1-d/p} + \tau), \quad (1.5)$$

where $\tau = \max_n \tau_n$. Because the upwind approximation employed in this paper corresponds to a one-sided difference approximation, the rate of convergence with respect to the spatial discretization is expected to be $O(h)$ at best. However, such a rate of convergence is not achieved in (1.5). That shortfall stems from the lack of regularity of solutions of a linear elliptic problem in a polygonal domain. Therefore, on considering (1.1) in a smooth domain, we can deduce (cf. Section 7) a refined estimate of the form

$$\sup_{0 \leq t_n \leq J} \|u(t_n) - u_h^n\|_{L^p(\Omega)} \leq C(h + \tau). \quad (1.6)$$

In order to derive (1.5) and (1.6), we formulate (1.1) and its finite-element approximation as single equations for u and u_h^n , respectively, and apply analytical (holomorphic) semigroup theory in Banach spaces. Actually, as proved in Crouzeix & Thomée (2001), if the triangulation is of acute type, the operator A_h , a finite-element approximation of $-\Delta + 1$ of the lumped mass type, becomes sectorial on a finite-element space $\mathcal{X}_{h,p}$ equipped with a modified L^p norm. In particular, $-A_h$ generates an analytic semigroup on $\mathcal{X}_{h,p}$. (The precise meaning of these symbols will be given in Section 4.) We then make use of Duhamel's principle, fractional powers of operators and the smoothing property of the semigroup. Lemma 4.4, which is an application of the Heinz–Krein inequality (Lemma A.1), also plays an important role. Although semigroup theory is somewhat abstract, several L^p estimates can be derived in quite a formal manner. Furthermore, our method of analysis is a discrete analogue of the standard approach for solving non-linear evolution problems. It also provides a theoretical framework for the approximation error analysis of some other non-linear evolution problems.

Before concluding this section, we briefly discuss some other results that are related to numerical methods for (1.1). Nakaguchi & Yagi (2002) presented finite-element/Runge–Kutta discretizations for the original Keller–Segel system without any numerical results. They also established error estimates in the $H^{1+\varepsilon}$ norm, $\varepsilon \in (0, 1/2)$, for a sufficiently small J , though they devoted little attention to conservation of the L^1 norm of approximate solutions. Marrocco (2003) discussed mixed finite-element approximations for (1.1) and offered various numerical examples, but a convergence analysis was not undertaken. Finally, as stated above, we presented finite-difference approximations to (1.1) enjoying discrete positivity and mass conservation properties in Saito & Suzuki (2005), although a convergence analysis was not performed in that study either.

The plan of this paper is as follows:

1. Introduction
2. Main results
3. Conservation laws (proof of Theorems 2.1 and 2.2)
4. Some preliminary results
5. Lemmas concerning b and b_h
6. Error estimates (proof of Theorem 2.3)
7. Remarks on error estimates of the order h
8. Numerical examples
- A. Proof of Lemma 4.4

Notation. We follow the notation of Adams & Fournier (2003). We put $W^{m,p} = W^{m,p}(\Omega)$, $H^m = W^{m,2}$, $L^p = L^p(\Omega)$, $\|\cdot\|_{m,p} = \|\cdot\|_{W^{m,p}}$, $\|\cdot\|_p = \|\cdot\|_{L^p}$ for $m \in \mathbb{N}$ and $p \in [1, \infty]$. The standard inner product in L^2 is denoted by $(\cdot, \cdot)$. We put, for $p \in [1, \infty)$,

$$\mathcal{W}_p = \left\{ v \in W^{2,p} \left| \frac{\partial v}{\partial \nu} = 0 \quad \text{on } \partial\Omega \right. \right\}.$$

We set $[a]_{\pm} = \max\{0, \pm a\}$ for $a \in \mathbb{R}$. The d -dimensional Lebesgue measure of $\mathcal{O} \subset \mathbb{R}^d$ is denoted by $\text{meas}(\mathcal{O}) = \text{meas}_d(\mathcal{O})$. For a Banach space X , its dual space is denoted as X^* . Generic positive constants depending on Ω are denoted as C , C' and so forth. In particular, C does not depend on the discretization parameters h and τ described below. If it is necessary to specify the dependence on other parameters, say α , β , then we write them as $C_{\alpha,\beta}$ or $C(\alpha, \beta)$. However, if the contribution of those parameters is not necessary for our argument, we omit indicating them. We shall use the same symbol I to indicate the identity operator on any space.

2. Main results

We assume that Ω is a convex polyhedral domain with the boundary $\partial\Omega$. In order to present our finite-element scheme, we introduce a weak formulation of (1.1). Since the bilinear form $(\nabla v, \nabla \chi) + (v, \chi)$ is continuous and coercive on $H^1 \times H^1$, for a given $f \in L^2$, there is a unique solution $v \in H^1$ of

$$(\nabla v, \nabla \chi) + (v, \chi) = (f, \chi) \quad \forall \chi \in H^1.$$

By elliptic regularity, we have $v \in \mathcal{W}_2$ and $\|v\|_{2,2} \leq C\|f\|_2$. We write $v = Gf$ to express this correspondence and call $G : L^2 \rightarrow \mathcal{W}_2 \subset L^2$ the Green operator associated with $-\Delta + 1$ on L^2 . We introduce a trilinear form b on $L^2 \times H^1 \times H^1$ defined as

$$b(w, u, \chi) = - \int_{\Omega} u \nabla G w \cdot \nabla \chi \, dx \quad (w \in L^2, u \in H^1, \chi \in H^1),$$

which is well-defined:

$$\begin{aligned} |b(w, u, \chi)| &\leq \|u\|_3 \|\nabla G w\|_6 \|\nabla \chi\|_2 && \text{(Hölder's inequality)} \\ &\leq \|u\|_{1,2} \|\nabla G w\|_{1,2} \|\nabla \chi\|_2 && \text{(Sobolev's inequality)} \\ &\leq \|u\|_{1,2} \|w\|_2 \|\nabla \chi\|_2 && \text{(elliptic regularity)} \end{aligned}$$

for $w \in L^2, u \in H^1, \chi \in H^1$.

A weak form of (1.1) is represented as follows: find $u \in C^1([0, J]; H^1)$ such that

$$\begin{cases} \left(\frac{du(t)}{dt}, \chi \right) + (k \nabla u(t), \nabla \chi) + \lambda b(u(t), u(t), \chi) = 0 & \forall \chi \in H^1, \forall t \in (0, J), \\ u(0) = u_0 \in H^1. \end{cases} \quad (2.1)$$

We proceed to the presentation of our finite-element approximation. Let $\{\mathcal{T}_h\} = \{\mathcal{T}_h\}_{h \downarrow 0}$ be a regular family of triangulations $\mathcal{T}_h$ of Ω :

1. $\mathcal{T}_h$ is a set of closed d -simplices (elements), and $\overline{\Omega} = \cup \{T | T \in \mathcal{T}_h\}$;
2. any two elements of $\mathcal{T}_h$ meet only in entire common faces or sides or in vertices;
3. there exists a positive constant γ_1 such that

$$h_T \leq \gamma_1 \rho_T \quad \forall T \in \mathcal{T}_h \in \{\mathcal{T}_h\}_h,$$

where

$$h_T = \text{diam}(T) \quad (\equiv \text{the diameter of } T),$$

$$\rho_T = \max\{\text{diam}(S) | S \text{ is a ball included in } T\}.$$

As the granularity parameter, we have employed $h = \max\{h_T | T \in \mathcal{T}_h\}$. Let $\mathcal{P}_h = \{P_i\}_{i=1}^N$ be the set of all vertices of $\mathcal{T}_h$, N being a positive integer. (Although N depends on h , we shall not explicitly express the dependence.) We divide $\mathcal{P}_h$ into two subsets $\{P_i\}_{i=1}^{N_I} \subset \Omega$ and $\{P_{N_I+i}\}_{i=1}^{N_B} \subset \partial\Omega$, where $N = N_I + N_B$. With P_i , we associate a function $\hat{\phi}_i \in C(\overline{\Omega})$ such that $\hat{\phi}_i$ is linear on each $T \in \mathcal{T}_h$ and $\hat{\phi}_i(P_j) = \delta_{ij}$, where δ_{ij} denotes Kronecker's delta. We define

$$X_h = \text{the vector space spanned by } \{\hat{\phi}_i\}_{i=1}^N$$

and regard it as a closed subspace of H^1 . We also consider the space X_h , which is equipped with the topology induced from L^2 , and denote it using the same symbol X_h . The barycentric domain D_i corresponding to P_i is defined as

$$D_i = \bigcup_{T \in \mathcal{S}_i} \{x \in T \mid \phi_j^T(x) \leq \phi_i^T(x) (P_j \in \mathcal{V}(T), P_j \neq P_i)\},$$

where $\mathcal{S}_i = \{S \in \mathcal{T}_h | P_i \in S\}$, $\mathcal{V}(T) = \{P_j \in \mathcal{P}_h | P_j \in T\}$ and $\{\phi_i^T\}_{i=1}^{d+1}$ are the barycentric coordinates of T with respect to P_i . Let $\bar{\phi}_i \in L^\infty$ be the characteristic function of D_i . We introduce a Hilbert space $\bar{X}_h \subset L^2$ spanned by $\{\bar{\phi}_i\}_{i=1}^N$. The operator $M_h: X_h \rightarrow \bar{X}_h$ is defined as

$$M_h v_h = \sum_{i=1}^N v_h(P_i) \bar{\phi}_i \quad (v_h \in X_h),$$

which is called the lumping operator. We put

$$(v_h, \chi_h)_h = (M_h v_h, M_h \chi_h) \quad (v_h, \chi_h \in X_h).$$

Thereby, $(\cdot, \cdot)_h^{1/2}$ is equivalent to $\|\cdot\|_2$ on X_h (see Section 4.2). The discrete Green operator $G_h: X_h \rightarrow X_h$ is defined as follows: For any $f_h \in X_h$, we consider the unique solution $v_h = G_h f_h \in X_h$ of

$$(\nabla v_h, \nabla \chi_h) + (v_h, \chi_h) = (f_h, \chi_h) \quad \forall \chi_h \in X_h.$$

To state our approximation of the trilinear form b , we need additional notation. Let $P_i, P_j \in \mathcal{P}_h$. If $P_i \neq P_j$ and they share an edge, we set

$$S_h^{ij} = \{T \in \mathcal{T}_h \mid P_i, P_j \in T\};$$

$$\Gamma_{ij} = \partial D_i \cap \partial D_j \text{ and}$$

$$v_{ij} = \text{the outer unit normal vector to } \Gamma_{ij} \text{ with respect to } D_i.$$

Otherwise, we set $S_h^{ij} = \emptyset$, $\Gamma_{ij} = \emptyset$ and $v_{ij} = 0$. The restrictions of Γ_{ij} and v_{ij} on $T \in S_h^{ij}$ are denoted by Γ_{ij}^T and v_{ij}^T , respectively. Then we introduce the functionals $\beta_{ij}^\pm$ on X_h as

$$\beta_{ij}^\pm(w_h) = \int_{\Gamma_{ij}} [\nabla G_h w_h \cdot v_{ij}]_\pm dS = \sum_{T \in S_h^{ij}} \text{meas}(\Gamma_{ij}^T) [(\nabla G_h w_h)_T \cdot v_{ij}^T]_\pm.$$

At this stage, writing

$$A_i = \{P_j \in \mathcal{P}_h \mid S_h^{ij} \neq \emptyset\},$$

we introduce a trilinear form b_h on $X_h \times X_h \times X_h$ by

$$b_h(w_h, u_h, \chi_h) = \sum_{i=1}^N \chi_h(P_i) \sum_{j \in A_i} \{u_h(P_i) \beta_{ij}^+(w_h) - u_h(P_j) \beta_{ij}^-(w_h)\} \quad (w_h, u_h, \chi_h \in X_h).$$

Although this is a direct application of the scheme of Baba & Tabata (1981), we briefly state its derivation in a formal manner:

$$\begin{aligned} b(w_h, u_h, \chi_h) &= \int_{\Omega} \nabla(u_h \nabla G_h w_h) \chi_h dx && \text{(integration by parts)} \\ &\approx \sum_{i=1}^N \chi_h(P_i) \int_{D_i} \nabla(u_h \nabla G_h w_h) dx && (\chi_h \approx M_h \chi_h) \\ &= \sum_{i=1}^N \chi_h(P_i) \sum_{j \in A_i} \int_{\Gamma_{ij}} u_h \nabla G_h w_h \cdot v_{ij} dS && \text{(divergence theorem)} \\ &\approx \sum_{i=1}^N \chi_h(P_i) \sum_{j \in A_i} \int_{\Gamma_{ij}} u_h \nabla G_h w_h \cdot v_{ij} dS && (G \approx G_h). \end{aligned}$$

Then, the integral on Γ_{ij} is approximated by considering the upwind nodal points as follows:

$$\int_{\Gamma_{ij}} u_h \nabla G_h w_h \cdot v_{ij} dS \approx u_h(P_i) \beta_{ij}^+(w_h) - u_h(P_j) \beta_{ij}^-(w_h).$$

Consequently, we obtain the trilinear form b_h .

As a discretization of the time variable, we take positive constants $\tau_1, \tau_2, \dots$, and put

$$t_n = \tau_1 + \dots + \tau_n \quad (n \geq 1).$$

We introduce the backward Euler difference quotient by setting

$$\partial_{\tau_n} w_h^n = \frac{w_h^n - w_h^{n-1}}{\tau_n} \quad (\{w_h^n\}_{n \geq 1} \subset X_h).$$

Now we can state our finite-element scheme for (2.1): Find $\{u_h^n\}_{n \geq 0} \subset X_h$ such that

$$\begin{cases} (\partial_{\tau_n} u_h^n, \chi_h)_h + (k \nabla u_h^n, \nabla \chi_h) + \lambda b_h(u_h^{n-1}, u_h^n, \chi_h) = 0 & \forall \chi_h \in X_h, \quad \forall n \geq 1, \\ u_h^0 = u_{0h}, \end{cases} \quad (2.2)$$

where $u_{0h} \in X_h$ is a suitable approximation of u_0 .

REMARK 2.1 The first equality of (2.2) includes two linear systems. Thus, at each time step, we initially solve a linear system for $G_h u_h^{n-1}$ and then we solve one for u_h^n . We note that $G_h u_h^n$ is an approximation of $v(t_n) = Gu(t_n)$.

REMARK 2.2 The semi-implicit time discretization in (2.2) is closely related to the reproduction of Lyapunov's property, which is another important feature of the system (2.1). For further details and other methods of time discretization, we refer to Saito & Suzuki (2005).

We are in a position to state our main theorems. The first one is related to the discrete version of (1.4).

THEOREM 2.1 (Conservation of total mass) Let $\{u_h^n\}_{n \geq 0} \subset X_h$ be a solution of (2.2). Then

$$(u_h^n, 1)_h = (u_{0h}, 1)_h \quad (2.3)$$

for $n \geq 0$.

The second theorem is related to the well-posedness of our scheme including the discrete version of (1.3). To state it, we put

$$\kappa_h = \min_{T \in \mathcal{T}_h} \kappa_T \quad (\kappa_T = \text{the minimal perpendicular length of } T)$$

and introduce the following conditions.

(H1) *Acuteness*. It is assumed that

$$\max\{\cos(\nabla \phi_i^T, \nabla \phi_j^T) | 1 \leq i, j \leq d+1\} \leq 0 \quad \forall T \in \mathcal{T}_h \in \{\mathcal{T}_h\},$$

where $\{\phi_i^T\}_{i=1}^{d+1}$ represent the barycentric coordinates of T with respect to the vertices of T .

REMARK 2.3 As stated in Section 3, (H1) guarantees the non-positivity of $(\nabla \hat{\phi}_i, \nabla \hat{\phi}_j)$ for $i \neq j$. See (3.2). For $d = 2$, (H1) is equivalent to a statement that each triangle of $\mathcal{T}_h$ is a right-angle or an acute triangle. For $d = 3$, (H1) is satisfied if, and only if, all angles spanned by two faces of each tetrahedron of $\mathcal{T}_h$ are less than or equal to $\pi/2$. Furthermore, for $d = 3$, (H1) is equivalent to the following condition: Given $T \in \mathcal{T}_h$, let $P \in T$ and $F \subset T$ be a vertex and the opposite face to P , respectively. Then the foot of the perpendicular from P to a plane S , including F , is always included in $\bar{F}$.

THEOREM 2.2 (Well-posedness and conservation of positivity) Let (H1) be satisfied. Suppose that $u_{0h} \in X_h$ is non-negative and is not identically constant. Take $\tau > 0$ and $\varepsilon \in (0, 1]$. Then, (2.2) with a

time step-size control

$$\tau_n = \min \left\{ \tau, \frac{3(d-1)\varepsilon\kappa_h}{\lambda d^2(d+1)\|\nabla G_h u_h^{n-1}\|_\infty} \right\}$$

admits a unique solution $\{u_h^n\}_{n \geq 0} \subset X_h$ such that $u_h^n > 0$ for $n \geq 1$.

Combining this with Theorem 2.1, we immediately obtain

COROLLARY 2.1 Let $\{u_h^n\}_{n \geq 0} \subset X_h$ be a solution of (2.2) as in Theorem 2.2. Then,

$$\|u_h^n\|_1 = \|u_{0h}\|_1$$

for $n \geq 0$.

REMARK 2.4 Because all norms are equivalent on X_h , we have

$$\begin{aligned} \|\nabla G_h u_h^{n-1}\|_\infty &\leq c_h \|G_h u_h^{n-1}\|_{1,2} \\ &\leq c_h \cdot C \|u_h^{n-1}\|_2 \\ &\leq c'_h \|u_h^{n-1}\|_1 = c'_h \|u_{0h}\|_1, \end{aligned}$$

where c_h, c'_h are positive constants depending on h . Therefore, there is a $c''_h > 0$ such that $\tau_n \geq \min\{\tau, c''_h\}$. Thus, τ_n never converges to zero as n increases, and therefore the algorithm always works. Consequently, u_h^n actually exists for all $n \geq 1$. Further, if we assume (H2) and (R) described below, then by making use of (4.2) and (4.31), we obtain a more explicit estimate as

$$\tau_n \geq \min \left\{ \tau, \frac{C\kappa_h h^{d(1-1/p)}}{\|u_{0h}\|_1} \right\}$$

with some $p \in (d, \mu)$ and $C = C(d, \Omega, \gamma_2) > 0$.

To state our convergence theorem, we require additional conditions:

(H2) *Quasiuniformity assumption.* There exists a positive constant γ_2 such that

$$\gamma_2 h \leq h_T \quad \forall T \in \mathcal{T}_h \in \{\mathcal{T}_h\}.$$

(R) *Elliptic regularity.* There exists $\mu \in (d, \infty)$ such that the following holds true: For any $p \in (1, \mu)$ and $f \in L^p(\Omega)$, the linear elliptic problem

$$-\Delta v + v = f \quad \text{in } \Omega, \quad \frac{\partial v}{\partial \nu} = 0 \quad \text{on } \partial\Omega \quad (2.4)$$

admits a unique solution $v \in \mathcal{W}_p$ that satisfies

$$\|v\|_{2,p} \leq C \|f\|_p \quad (2.5)$$

with a constant $C = C(p, \Omega) > 0$.

REMARK 2.5 When $\Omega \subset \mathbb{R}^2$ is a convex polygon, (R) is always satisfied. On the other hand, when $\Omega \subset \mathbb{R}^3$ is a convex polyhedron, it is satisfied, if all edges and all vertices of Ω are sufficiently small that they do not produce singularities. See, for more complete descriptions, Theorems 8.2.1.2 and 8.2.2.8 of Grisvard (1985).

REMARK 2.6 Under condition (R), for any $p \in (1, \mu)$, we introduce an operator $G_p: L^p \rightarrow \mathcal{W}_p$ defined by $v = G_p f$, where v is the unique solution of (2.4) for a given $f \in L^p$. Then the operator G should be understood precisely as $G = G_p$ with some $p \in (1, \mu)$. If we do not assume (R), then we understand that $G = G_2$.

Because $\partial\Omega$ is not smooth, we cannot treat classical solutions. Instead, we consider L^p solutions, which are well discussed in Suzuki & Senba (2004) and Suzuki (2005). Thus, we make the following regularity assumption on a solution u of (2.1):

$$u \in C([0, J]: \mathcal{W}_p), \quad u' \in C([0, J]: W^{1,p}) \cap C^\sigma([0, J]: L^p) \quad (2.6)$$

for some $p \geq 2$, $\sigma \in (0, 1]$, and put

$$\alpha_{1,p} = \sup_{t \in [0, J]} \|u(t)\|_{2,p}, \quad \alpha_{2,p} = \sup_{t \in [0, J]} \|u'(t)\|_{1,p}, \quad \alpha_{3,p} = \sup_{t, s \in [0, J]} \frac{\|u'(t) - u'(s)\|_p}{|t - s|^\sigma},$$

where $u' = du/dt$.

THEOREM 2.3 (Error estimates) Suppose that (H1), (H2) and (R) are satisfied. Assume that (2.1) admits a unique solution u satisfying (2.6) for some $p \in (d, \mu)$ and $\sigma \in (0, 1]$. Moreover, let $u_{0h} \in X_h$ be chosen as

$$\|u_0 - u_{0h}\|_p \leq \alpha_{0,p} h^{1-d/p}, \quad (2.7)$$

with some $\alpha_{0,p} = \alpha_{0,p}(u_0) > 0$. Then, there exist positive constants h_0, τ_0 depending on $\Omega, J, k, \lambda, p, \sigma, \gamma_i$'s and $\alpha_{i,p}$'s such that we have the error estimates

$$\sup_{0 \leq n \leq l} \|u(t_n) - u_h^n\|_p \leq C_1(h^{1-d/p} + \tau^\sigma), \quad (2.8)$$

$$\sup_{0 \leq n \leq l} \|Gu(t_n) - G_h u_h^n\|_{1,\infty} \leq C_2(h^{1-d/p} + \tau^\sigma) \quad (2.9)$$

for $h \in (0, h_0)$ and $\tau \in (0, \tau_0)$, where $l = l(\tau, h) = \max\{n \in \mathbb{N} \mid t_n < J\}$, $\{u_h^n\}_{n \geq 0} \subset X_h$ is the solution of (2.2) as in Theorem 2.2. Furthermore, the constants C_1 and C_2 can be taken as

$$C_i = C(J+1)(\alpha_{0,p} + \alpha_{1,p}^2 + \alpha_{2,p}^2 + \alpha_{3,p}) \exp[C'(1 + \alpha_{1,p}^2)J], \quad i = 1, 2,$$

where C and C' are positive constants that depend only on $\Omega, k, \lambda, \gamma_i$'s, h_0 and τ_0 .

3. Conservation laws (proof of Theorems 2.1 and 2.2)

We recall some standard results. Put

$$m_i = \text{meas}(D_i),$$

$$\kappa_i = \min\{\kappa_T \mid P_i \in T, T \in \mathcal{T}_h\},$$

$$a_{ij} = (\nabla \hat{\phi}_j, \nabla \hat{\phi}_i),$$

$$b_{ij}(w_h) = b_h(w_h, \hat{\phi}_j, \hat{\phi}_i) \quad (w_h \in X_h)$$

for $1 \leq i, j \leq N$.

We have

$$\sum_{i=1}^N a_{ij} = \sum_{i=1}^N a_{ji} = 0. \quad (3.1)$$

Under Assumption (H1), we have (cf. Fujii, 1973)

$$a_{ij} \leq 0 \quad (i \neq j), \quad 0 < \frac{a_{ii}}{m_i} \leq \frac{d+1}{\kappa_i^2}. \quad (3.2)$$

Moreover, under (H1), we readily verify that the $N \times N$ matrix $\mathbf{A} = [a_{ij}]$ is irreducible. As stated in Lemma 2.1 of Baba & Tabata (1981), for $w_h \in X_h$, we have

$$\sum_{i=1}^N b_{ij}(w_h) = 0, \quad (3.3)$$

$$b_{ij}(w_h) \leq 0 \quad (i \neq j), \quad 0 \leq \frac{b_{ii}(w_h)}{m_i} \leq \frac{2d}{\kappa_i} \|\nabla G_h w_h\|_\infty. \quad (3.4)$$

Furthermore, for $w_h \in X_h$ and $1 \leq i, j \leq N$, we set

$$\beta_{ij}(w_h) = \beta_{ij}^+(w_h) - \beta_{ij}^-(w_h). \quad (3.5)$$

Then, we easily obtain $\beta_{ij}^\pm(w_h) = \beta_{ji}^\mp(w_h)$, $\beta_{ij}(w_h) + \beta_{ji}(w_h) = 0$ and

$$|\beta_{ij}(w_h)| \leq \text{meas}(\Gamma_{ij}) \|\nabla G_h w_h\|_\infty. \quad (3.6)$$

Now we can state the following proof.

Proof of Theorem 2.1. Substituting $\chi_h = \hat{\phi}_i$ into (2.2), we obtain

$$m_i \frac{u_h^n(P_i) - u_h^{n-1}(P_i)}{\tau_n} = - \sum_{j=1}^N \{ka_{ij} + \lambda b_{ij}(u_h^{n-1})\} u_h^n(P_j).$$

Summing this with respect to i from 1 to N and using (3.1) and (3.3), we obtain

$$\frac{1}{\tau_n} \left(\sum_{i=1}^N u_h^n(P_i) m_i - \sum_{i=1}^N u_h^{n-1}(P_i) m_i \right) = - \sum_{j=1}^N \sum_{i=1}^N \{ka_{ij} + \lambda b_{ij}(u_h^{n-1})\} u_h^n(P_j) = 0,$$

which implies (2.3). $\square$

Proof of Theorem 2.2. It suffices to verify the following fact: For a given $0 \leq w_h \in X_h$ and $\delta > 0$ satisfying

$$\delta \leq \frac{3(d-1)\kappa_h}{\lambda d^2(d+1) \|\nabla G_h w_h\|_\infty}, \quad (3.7)$$

there is a unique $0 < u_h \in X_h$ such that

$$\left(\frac{u_h - w_h}{\delta}, \chi_h \right)_h + (k \nabla u_h, \nabla \chi_h) + \lambda b_h(w_h, u_h, \chi_h) = 0 \quad \forall \chi_h \in X_h. \quad (3.8)$$

As before, (3.8) implies that

$$\frac{u_h(P_i) - w_h(P_i)}{\delta} + \sum_{j=1}^N \frac{ka_{ij} + \lambda b_{ij}(w_h)}{m_i} u_h(P_j) = 0.$$

This is written equivalently as

$$\mathbf{C}\mathbf{u} = \mathbf{w}, \quad (3.9)$$

where $\mathbf{u} = (u_h(P_1), \dots, u_h(P_N))^T$, $\mathbf{w} = (w_h(P_1), \dots, w_h(P_N))^T$ and $\mathbf{C} = [c_{ij}]$ with

$$\begin{aligned} c_{ii} &= 1 + \frac{\delta}{m_i}(ka_{ii} + \lambda b_{ii}(w_h)) \quad (1 \leq i \leq N), \\ c_{ij} &= \frac{\delta}{m_i}(ka_{ij} + \lambda b_{ij}(w_h)) \quad (1 \leq i, j \leq N, i \neq j). \end{aligned}$$

If we can show that

$$c_{ij} \leq 0 \quad (i \neq j), \quad c_{ii} > 0; \quad (3.10)$$

$$\mathbf{C} \text{ is irreducibly diagonally dominant,} \quad (3.11)$$

then we apply Corollary 3.20 of Varga (2000) and obtain that $\mathbf{C}$ is regular and that every entry of $\mathbf{C}^{-1}$ is positive. These yield the desired fact.

Now we verify (3.10) and (3.11). First (3.10) is a direct consequence of (3.2) and (3.4). In order to check (3.11), let $1 \leq i, j \leq N$ with $i \neq j$. We have by (3.1) that

$$\begin{aligned} |c_{ii}| - \sum_{j \neq i} |c_{ij}| &= 1 + \frac{\delta}{m_i} \sum_{j=1}^N (ka_{ij} + \lambda b_{ij}(w_h)) \\ &= 1 + \frac{\delta \lambda}{m_i} \sum_{j=1}^N b_{ij}(w_h). \end{aligned} \quad (3.12)$$

Because

$$b_{ij}(w_h) = - \sum_{k \in A_i} \beta_{ik}^-(w_h) \delta_{jk} = \begin{cases} -\beta_{ij}^-(w_h) & (j \in A_i) \\ 0 & (j \notin A_i), \end{cases}$$

for $i \neq j$, we obtain, by (3.3), (3.5) and (3.6), that

$$\begin{aligned} \sum_{j=1}^N b_{ij}(w_h) &= b_{ii}(w_h) + \sum_{j \neq i} b_{ij}(w_h) = - \sum_{j \in A_i} b_{ji}(w_h) + \sum_{j \in A_i} b_{ij}(w_h) \\ &= \sum_{j \in A_i} [\beta_{ji}^-(w_h) - \beta_{ij}^-(w_h)] = \sum_{j \in A_i} [\beta_{ij}^+(w_h) - \beta_{ij}^-(w_h)] \\ &\geq -\|\nabla G_h w_h\|_\infty \sum_{j \in A_i} \text{meas}(I_{ij}). \end{aligned} \quad (3.13)$$

We recall (cf. Lemma 2.1 of Baba & Tabata, 1981) that there is $\zeta_i \in (0, 1]$ such that

$$\frac{1}{m_i} \sum_{j \in \mathcal{A}_i} \text{meas}(\Gamma_{ij}) \leq \zeta_i \frac{d^2(d+1)}{3(d-1)\kappa_i}.$$

In particular, we can take $\zeta_i = 1$ for $1 \leq i \leq N_I$ and $\zeta_i \in (0, 1)$ for $N_I + 1 \leq i \leq N$. Hence, by (3.7), (3.12) and (3.13),

$$|c_{ii}| - \sum_{j \neq i}^N |c_{ij}| \begin{cases} \geq 0 & (1 \leq i \leq N_I), \\ > 0 & (N_I + 1 \leq i \leq N). \end{cases}$$

This implies that $\mathbf{C}$ is diagonally dominant with strict inequalities holding for $N_I + 1 \leq i \leq N$. Finally, we verify the irreducibility. In view of (3.2) and (3.4), $a_{ij} < 0$ implies $c_{ij} < 0$ for $i \neq j$. Hence, the irreducibility of $\mathbf{C}$ is reduced to that of $\mathbf{A} = [a_{ij}]$. The proof is completed. $\square$

4. Some preliminary results

In this section, we collect some preliminary results, which will be useful in subsequent sections.

4.1 Sobolev and inverse inequalities

We shall make use of

$$\|v\|_\infty \leq C \|v\|_{1,p} \quad (p \in (d, \infty], v \in W^{1,p}). \quad (4.1)$$

In what follows, we often treat functions of $W^{1,p}$, $p > d$, as those of $C(\overline{\Omega})$. Next, let $T \in \mathcal{T}_h$. Assumption (H2) implies the inverse inequality

$$\|v_h\|_{W^{l,p}(T)} \leq Ch_T^{m-l+\min\{0, \frac{d}{p}-\frac{d}{q}\}} \|v_h\|_{W^{m,q}(T)} \quad (v_h \in X_h), \quad (4.2)$$

where $p, q \in [1, \infty]$ and $0 \leq m \leq l$ are integers. It is readily verifiable that

$$\max_{x,y \in T} |v_h(x) - v_h(y)| \leq Ch_T^{1-d/p} \|\nabla v_h\|_{L^p(T)} \quad (p \in [1, \infty], v_h \in X_h). \quad (4.3)$$

These facts are explained more thoroughly in Adams & Fournier (2003), Brenner & Scott (2002) and Ciarlet (1978).

4.2 Some auxiliary operators

The Lagrange interpolation operator $\pi_h: C(\overline{\Omega}) \rightarrow X_h$ is defined by $\pi_h v(P_i) = v(P_i)$ for all $P_i \in \mathcal{P}_h$. For $T \in \mathcal{T}_h$, we have

$$\|\pi_h v - v\|_{L^p(T)} + h \|\nabla(\pi_h v - v)\|_{L^p(T)} \leq Ch^2 \|v\|_{W^{2,p}(T)} \quad (p \in (d/2, \infty], v \in W^{2,p}). \quad (4.4)$$

Moreover, under (H2), we have (cf. Theorem 4.4.20 of Brenner & Scott, 2002)

$$\|\pi_h v - v\|_{L^\infty(T)} + h \|\nabla(\pi_h v - v)\|_{L^\infty(T)} \leq Ch^{2-d/p} \|v\|_{W^{2,p}(T)} \quad (p \in (d, \infty], v \in W^{2,p}). \quad (4.5)$$

We frequently use the L^2 and H^1 projection operators $P_h: L^2 \rightarrow X_h$ and $R_h: H^1 \rightarrow X_h$, which are defined as

$$(P_h v - v, \chi_h) = 0 \quad \forall \chi_h \in X_h, \quad (4.6)$$

$$(\nabla R_h v - \nabla v, \nabla \chi_h) + (R_h v - v, \chi_h) = 0 \quad \forall \chi_h \in X_h. \quad (4.7)$$

LEMMA 4.1 Under (H2), we have

$$\|P_h v\|_p \leq C \|v\|_p \quad (p \in [1, \infty], v \in L^p), \quad (4.8)$$

$$\|P_h v\|_{1,p} \leq C \|v\|_{1,p} \quad (p \in [1, \infty], v \in W^{1,p}), \quad (4.9)$$

$$\|P_h v - v\|_p \leq C h^2 \|v\|_{2,p} \quad (p \in (d/2, \infty], v \in W^{2,p}). \quad (4.10)$$

Proof. Inequalities (4.8) and (4.9) are attributed to Douglas *et al.* (1975) and Crouzeix & Thomée (1987). Inequality (4.10) follows from (4.4) and (4.8). $\square$

LEMMA 4.2 Let (H2) and (R) be satisfied. Then we have

$$\|R_h v\|_{1,p} \leq C \|v\|_{1,p} \quad (p \in (1, \infty], v \in W^{1,p}), \quad (4.11)$$

$$\|R_h v - v\|_{1,p} \leq C h \|v\|_{2,p} \quad (p \in (1, \infty], v \in W^{2,p}), \quad (4.12)$$

$$\|R_h v - v\|_p \leq C h^2 \|v\|_{2,p} \quad (p \in (\mu/(\mu-1), \infty), v \in W^{2,p}), \quad (4.13)$$

$$\|R_h v - v\|_{1,\infty} \leq C h^{1-d/p} \|v\|_{2,p} \quad (p \in (d, \infty], v \in W^{2,p}). \quad (4.14)$$

Proof. The derivation of (4.11)–(4.13) is the same as that shown in Chapter 8 of Brenner & Scott (2002). Therein, the case of the Dirichlet boundary condition was considered explicitly. To show (4.14), we note that $\|R_h v - v\|_{1,\infty} \leq \|R_h v - \pi_h v\|_{1,\infty} + \|\pi_h v - v\|_{1,\infty} \leq C \|\pi_h v - v\|_{1,\infty}$ for $v \in W^{1,\infty}$ by (4.11). Hence, (4.14) follows from (4.5). $\square$

Let M_h^* be the adjoint operator of M_h in L^2 , and set

$$K_h = M_h^* M_h. \quad (4.15)$$

Thereby, we have

$$C \|v_h\|_p \leq \|M_h v_h\|_p \leq C' \|v_h\|_p \quad (p \in [1, \infty], v_h \in X_h). \quad (4.16)$$

Moreover, if (H2) is satisfied, then

$$C \|v_h\|_p \leq \|K_h v_h\|_p \leq C' \|v_h\|_p \quad (p \in [1, \infty], v_h \in X_h), \quad (4.17)$$

and

$$\|M_h v_h - v_h\|_p \leq C h \|\nabla v_h\|_p \quad (p \in [1, \infty], v_h \in X_h). \quad (4.18)$$

See Chapter 5 of Fujita *et al.* (2001) for (4.16), (4.17), and Baba & Tabata (1981) for (4.18). Inequality (4.17) is valid for $p = 2$ without (H2).

REMARK 4.1 Most results that are recalled in this paragraph are valid for a sufficiently small h , or more precisely, for $h \in (0, h_*)$ with some suitable $h_* = h_*(\Omega, \gamma_1, \gamma_2) > 0$. However, in order to simplify the presentation, we omit its precise definition here. Similarly, in what follows, h will be taken from $(0, h_*)$ with some suitable $h_* = h_*(\Omega, \gamma_1, \gamma_2) > 0$.

4.3 Discrete Laplace operator

Throughout this paragraph, we suppose that (R) is satisfied. We recall the operator $G_p: L^p \rightarrow \mathcal{W}_p$ defined by $v = G_p f$, where v is the unique solution of (2.4) for a given $f \in L^p$. We set $A_p = G_p^{-1}$, which is nothing but the L^p realization of $-\Delta + I$ with the Neumann boundary condition

$$\mathcal{D}(A_p) = \mathcal{W}_p, \quad A_p v = -\Delta v + v \quad (v \in \mathcal{D}(A_p)). \quad (4.19)$$

We simply write G and A to express G_p and A_p , respectively, if there is no possibility of confusion (see Remark 2.6). The closed linear operator A is sectorial in L^p . Therefore, its fractional powers A^α , $\alpha \in (0, 1)$, are defined in a natural way, and $-A$ generates the analytic semigroup $\{e^{-tA}\}_{t \geq 0}$. From the general theory, we have

$$\mathcal{D}(A^{1/2}) = W^{1,p}, \quad (4.20)$$

$$\|A^\alpha e^{-tA}\|_p \leq C t^{-\alpha} \quad (t > 0, \alpha \in (0, 1)). \quad (4.21)$$

For these facts, see Henry (1981) and Tanabe (1979). We introduce the operators L_h and A_h of $X_h \rightarrow X_h$ defined as

$$\begin{aligned} L_h u_h &= f_h \Leftrightarrow (\nabla u_h, \nabla \chi_h) + (u_h, \chi_h) = (f_h, \chi_h) \quad \forall \chi_h \in X_h, \\ A_h u_h &= f_h \Leftrightarrow (\nabla u_h, \nabla \chi_h) + (u_h, \chi_h)_h = (f_h, \chi_h)_h \quad \forall \chi_h \in X_h. \end{aligned}$$

Obviously, we have $L_h = G_h^{-1}$ and

$$A_h = K_h^{-1} L_h. \quad (4.22)$$

Moreover,

$$L_h R_h v = P_h A v \quad (v \in \mathcal{D}(A)), \quad (4.23)$$

or equivalently,

$$R_h G w = G_h P_h w \quad (w \in L^p). \quad (4.24)$$

We collect some operator theoretical properties of A_h . To this end, in this paragraph, we regard any function space as a complex valued one, and propose a re-definition:

$$(u, v) = \int_{\Omega} u(x) \overline{v(x)} \, dx \quad \left(u \in L^p, v \in L^q, \frac{1}{p} + \frac{1}{q} = 1 \right).$$

For $p \in [1, \infty)$, we introduce the discrete L^p norm

$$\|v_h\|_{h,p} = \left(\int_{\Omega} M_h \pi_h |v_h(x)|^p \, dx \right)^{1/p} \quad (v_h \in X_h).$$

It is readily verifiable that

$$C \|v_h\|_{h,p} \leq \|v_h\|_p \leq C' \|v_h\|_{h,p} \quad (v_h \in X_h), \quad (4.25)$$

$$|(v_h, \chi_h)_h| \leq \|v_h\|_{h,p} \|\chi_h\|_{h,q} \quad \left(v_h, \chi_h \in X_h, \frac{1}{p} + \frac{1}{q} = 1 \right). \quad (4.26)$$

We regard X_h as a Banach space equipped with the norm $\|\cdot\|_{h,p}$ and indicate it by $\mathcal{X}_{h,p}$. In particular, $\mathcal{X}_{h,2}$ forms a Hilbert space with respect to the inner product $(\cdot, \cdot)_h$. Furthermore, the operator norm in $\mathcal{X}_{h,p}$ is denoted by the same symbol $\|\cdot\|_{h,p}$. For instance,

$$\|A_h\|_{h,p} = \sup_{v_h \in X_h} \frac{\|A_h v_h\|_{h,p}}{\|v_h\|_{h,p}}.$$

LEMMA 4.3 Let $p \in (1, \infty)$, and suppose that (H1) is satisfied. Then,

- (i) A_h is sectorial in $\mathcal{X}_{h,p}$, and its fractional powers A_h^α , $\alpha \in (0, 1)$, are defined.
- (ii) A_h and A_h^α , $\alpha \in (0, 1)$, are positive and self-adjoint in $\mathcal{X}_{h,2}$.
- (iii) if (H2) is also satisfied, for any $\theta \in (0, 1)$ and $\{\tau_j\}_{j=1}^n$, $\tau_j > 0$, we have

$$\|r(\tau_n A_h) \cdots r(\tau_1 A_h) A_h^\theta\|_{h,p} \leq C_\theta (\tau_n + \cdots + \tau_1)^{-\theta}, \quad (4.27)$$

where $r(\tau_j A_h) = (I + \tau_j A_h)^{-1}$.

Proof. (i) See Crouzeix & Thomée (2001). (ii) is easy to verify (see also Saito, 2004a). (iii) follows from Theorem 4 of Saito (2004b). $\square$

REMARK 4.2 Since A_h is a linear operator on a finite-dimensional space, (i) and (ii) hold simultaneously. Such a fact generally fails in infinite-dimensional spaces. In fact, A_2 is self-adjoint in L^2 , whereas A_p is not self-adjoint for $p \neq 2$.

REMARK 4.3 Inequality (4.27) is equivalently written as

$$\|A_h^\theta r(\tau_n A_h) \cdots r(\tau_1 A_h)\|_{h,p} \leq C_\theta (\tau_n + \cdots + \tau_1)^{-\theta}, \quad (4.28)$$

which is a discrete analogue of (4.21).

The following lemma, which is a discrete analogue of (4.20), will be proved in Appendix A.

LEMMA 4.4 Suppose that (H1), (H2) and (R) are satisfied. Then,

$$\|v_h\|_{1,p} \leq C \|A_h^\theta v_h\|_{h,p} \quad (v_h \in X_h) \quad (4.29)$$

for $p \in (\mu/(\mu - 1), \mu)$ and $\theta \in (1/2, 1]$. (When $p = 2$, we can take $\theta = 1/2$ without (R).)

REMARK 4.4 We do not know whether the restriction on p and the condition (R) are necessary or not for (4.29) to hold. See also Remark A.1. We note that these assumptions produce no obstruction in our arguments. Actually, in Section 6, we suppose that $p \in (d, \mu)$. Hence, the conjugate exponent of p is in the interval $(\mu/(\mu - 1), d/(d - 1))$.

REMARK 4.5 As a readily obtainable consequence of (4.29), we deduce that if (H1), (H2) and (R) are satisfied, then

$$\|\nabla A_h^{-\theta} v_h\|_p \leq \|A_h^{-\theta} v_h\|_{1,p} \leq C \|v_h\|_{h,p} \quad (v_h \in X_h) \quad (4.30)$$

for $p \in (\mu/(\mu - 1), \mu)$ and $\theta \in (1/2, 1]$.

LEMMA 4.5 Suppose that (H2) and (R) are satisfied. Then, for $p \in (d, \mu)$,

$$\|v_h\|_{1,\infty} \leq C \|L_h v_h\|_p \quad (v_h \in X_h). \quad (4.31)$$

Proof. By virtue of (4.11), (4.1) and (2.5), we have

$$\|R_h v\|_{1,\infty} \leq C \|v\|_{1,\infty} \leq C \|v\|_{2,p} \leq C \|Av\|_p$$

for $v \in \mathcal{D}(A)$. Let $v_h \in X_h$. Choosing $v = A^{-1}v_h$, we obtain by (4.23)

$$\|L_h^{-1}v_h\|_{1,\infty} \leq C \|v_h\|_p,$$

which implies (4.31). $\square$

4.4 Dual norms

Let $p, q \in (1, \infty)$ such that $\frac{1}{p} + \frac{1}{q} = 1$. It is evident that $\mathcal{X}_{h,p} = \mathcal{X}_{h,q}^*$. Therefore,

$$\|v_h\|_{h,p} \leq C \sup_{\chi_h \in X_h} \frac{(v_h, \chi_h)_h}{\|\chi_h\|_{h,q}} \quad (v_h \in X_h). \quad (4.32)$$

On the other hand, we consider the space X_h equipped with $\|\cdot\|_p$ and regard it as a Banach space $X_{h,p}$. Then, we have $X_{h,p} = X_{h,q}^*$ and

$$\|v_h\|_p \leq C \sup_{\chi_h \in X_h} \frac{(v_h, \chi_h)}{\|\chi_h\|_q} \quad (v_h \in X_h). \quad (4.33)$$

4.5 A lemma concerning K_h

LEMMA 4.6 Let (H1), (H2) and (R) be satisfied. Then,

$$\|A_h^{-\theta}(K_h^{-1} - I)v_h\|_{h,p} \leq Ch^2 \|\nabla v_h\|_p \quad (4.34)$$

for $p \in (\mu/(\mu - 1), \mu)$, $\theta \in (1/2, 1]$ and $v_h \in X_h$.

Proof. In the same way as in the proof of Lemma 15.1 of Thomée (1997), we have

$$|((K_h - I)v_h, \chi_h)| = |(M_h v_h, M_h \chi_h) - (v_h, \chi_h)| \leq Ch^2 \|\nabla v_h\|_p \|\nabla \chi_h\|_q \quad (4.35)$$

for $v_h, \chi_h \in X_h$ and $\frac{1}{p} + \frac{1}{q} = 1$. Hence, by (4.25) and (4.30),

$$\begin{aligned} |(A_h^{-\theta}(K_h^{-1} - I)v_h, \chi_h)_h| &= |((K_h^{-1} - I)v_h, A_h^{-\theta} \chi_h)_h| \\ &= |((I - K_h)v_h, A_h^{-\theta} \chi_h)| \\ &\leq Ch^2 \|\nabla v_h\|_p \|\nabla A_h^{-\theta} \chi_h\|_q \\ &\leq Ch^2 \|\nabla v_h\|_p \|\chi_h\|_{h,q} \end{aligned}$$

for $v_h, \chi_h \in X_h$. This, together with (4.32), yields (4.34). $\square$

4.6 Discrete inequality of Volterra type

LEMMA 4.7 Taking positive constants $\tau_1, \dots, \tau_l$ ($l \in \mathbb{N}$), putting $t_n = \tau_1 + \dots + \tau_n$ for $1 \leq n \leq l$ and $t_0 = 0$, suppose that a sequence $\{z_n\}_{n=0}^l \subset \mathbb{R}$ satisfies

$$0 < z_n \leq c_1 + c_2 \sum_{j=1}^n \frac{\tau_j}{(t_n - t_{j-1})^r} (z_{j-1} + z_j) \quad (1 \leq n \leq l), \quad (4.36)$$

where c_1, c_2 and $r \in (0, 1)$ are positive constants. Then we have

$$z_n \leq c_1 c_3 \exp \left(c_4 c_2^{\frac{1}{1-r}} t_n \right) \quad (0 \leq n \leq l),$$

where c_3 and c_4 are positive constants depending only on r .

Proof. Setting $Z = \max_{0 \leq n \leq l} z_n$, we have by (4.36) that

$$z_n \leq c_1 + 2Zc_2 \int_0^{t_n} (t_n - s)^{-r} ds = c_1 + 2Zc_2 t_n^{1-r} B(1-r, 1),$$

where $B(\cdot, \cdot)$ denotes Euler's B -function. The rest of the proof is exactly the same as that of Lemma 6.5 of Okamoto (1982). $\square$

5. Lemmas concerning b and b_h

LEMMA 5.1 Let (R) be satisfied. Let $p \in (d, \mu)$ and $\frac{1}{p} + \frac{1}{q} = 1$. Then,

$$|b(w - P_h w, u, \chi_h)| \leq Ch^2 \|w\|_{2,p} \|u\|_\infty \|\nabla \chi_h\|_q \quad (5.1)$$

for $w \in W^{2,p}$, $u \in L^\infty$, $\chi_h \in X_h$, and

$$|b(w, u, \chi_h - M_h \chi_h)| \leq Ch \|w\|_p \|u\|_{1,p} \|\nabla \chi_h\|_q \quad (5.2)$$

for $w \in L^p$, $u \in W^{1,p}$, $\chi_h \in X_h$.

Proof. Let $w \in W^{2,p}$, $u \in L^\infty$ and $\chi_h \in X_h$. Then, by (2.5) and (4.10),

$$\begin{aligned} |b(w - P_h w, u, \chi_h)| &\leq \|u\|_\infty \|\nabla G(w - P_h w)\|_p \|\nabla \chi_h\|_q \\ &\leq \|u\|_\infty \|G(w - P_h w)\|_{2,p} \|\nabla \chi_h\|_q \\ &\leq \|u\|_\infty \|w - P_h w\|_p \|\nabla \chi_h\|_q \\ &\leq \|u\|_\infty Ch^2 \|w\|_{2,p} \|\nabla \chi_h\|_q \end{aligned}$$

which implies (5.1). On the other hand, by integration by parts,

$$b(w, u, \chi_h - M_h \chi_h) = \int_\Omega \nabla (u \nabla G w) \cdot (\chi_h - M_h \chi_h) dx$$

for $w \in L^p$, $u \in W^{1,p}$ and $\chi_h \in X_h$. Since $p > d$, we can perform an estimation using (2.5):

$$\|\nabla \cdot (u \nabla G w)\|_p \leq C \|u\|_{1,p} \|\nabla G w\|_{1,p} \leq C \|u\|_{1,p} \|w\|_p.$$

Combining these with (4.18), we obtain (5.2). $\square$

Next we derive some estimates for b_h . Putting

$$\Gamma_h = \{\Gamma_{ij} = \partial D_i \cap \partial D_j | 1 \leq i, j \leq N\}, \quad (5.3)$$

we observe that, for $w_h, u_h, \chi_h \in X_h$,

$$\begin{aligned} b_h(w_h, u_h, \chi_h) &= \sum_{\Gamma_{ij} \in \Gamma_h} [\chi_h(P_i) - \chi_h(P_j)] [\beta_{ij}^+(w_h) u_h(P_i) - \beta_{ij}^-(w_h) u_h(P_j)] \\ &= \sum_{\Gamma_{ij} \in \Gamma_h} [\chi_h(P_i) - \chi_h(P_j)] [\sigma_{ij}^+(w_h) u_h(P_i) - \sigma_{ij}^-(w_h) u_h(P_j)] \beta_{ij}(w_h), \end{aligned} \quad (5.4)$$

where

$$\sigma_{ij}^+(w_h) = \operatorname{sgn} \beta_{ij}^+(w_h), \quad \sigma_{ij}^-(w_h) = 1 - \sigma_{ij}^-(w_h). \quad (5.5)$$

LEMMA 5.2 Let $p, q \in (1, \infty)$ with $\frac{1}{p} + \frac{1}{q} = 1$. Then, under (H2),

$$|b_h(w_h, u_h, \chi_h)| \leq \begin{cases} C \|\nabla G_h w_h\|_p \|u_h\|_\infty \|\nabla \chi_h\|_q, \\ C \|\nabla G_h w_h\|_\infty \|u_h\|_p \|\nabla \chi_h\|_q \end{cases} \quad (5.6)$$

for $w_h, u_h, \chi_h \in X_h$.

Proof. For $1 \leq i, j \leq N$, we put

$$h_{T_{ij}} = \max\{h_T | T \in S_h^{ij}\}. \quad (5.7)$$

(If $S_h^{ij} = \emptyset$, then $h_{T_{ij}} = 0$.) Let $w_h, u_h, \chi_h \in X_h$. By (5.4) and (4.3), we can deduce that

$$\begin{aligned} |b_h(w_h, u_h, \chi_h)| &\leq \sum_{\Gamma_{ij} \in \Gamma_h} |\chi_h(P_i) - \chi_h(P_j)| \cdot |\sigma_{ij}^+(w_h) u_h(P_i) - \sigma_{ij}^-(w_h) u_h(P_j)| \cdot |\beta_{ij}(w_h)| \\ &\leq \sum_{\Gamma_{ij} \in \Gamma_h} C h_{T_{ij}}^{1-\frac{d}{q}} \|\nabla \chi_h\|_{L^q(T_{ij})} \|u_h\|_{L^\infty(\Gamma_{ij})} \|\nabla G_h w_h\|_{L^\infty(\Gamma_{ij})} \operatorname{meas}(\Gamma_{ij}) \\ &\leq C \|u_h\|_\infty \sum_{\Gamma_{ij} \in \Gamma_h} h_{T_{ij}}^{1-\frac{d}{q}} \|\nabla \chi_h\|_{L^q(T_{ij})} h_{T_{ij}}^{-\frac{d}{p}} \|\nabla G_h w_h\|_{L^p(T_{ij})} h_{T_{ij}}^{d-1} \\ &\leq C \|u_h\|_\infty \sum_{T \in \mathcal{T}_h} \|\nabla \chi_h\|_{L^q(T)} \|\nabla G_h w_h\|_{L^p(T)} \\ &\leq C \|u_h\|_\infty \|\nabla \chi_h\|_q \|\nabla G_h w_h\|_p. \end{aligned}$$

Similarly, we obtain

$$\begin{aligned} |b_h(w_h, u_h, \chi_h)| &\leq C \|\nabla G_h w_h\|_\infty \sum_{T \in \mathcal{T}_h} \|\nabla \chi_h\|_{L^q(T)} \|u_h\|_{L^p(T)} \\ &\leq C \|\nabla G_h w_h\|_\infty \|u_h\|_p \|\nabla \chi_h\|_q, \end{aligned}$$

which completes the proof. $\square$

LEMMA 5.3 Suppose that (H2) and (R) are satisfied. Let $p, q \in (1, \infty)$ with $\frac{1}{p} + \frac{1}{q} = 1$, and let $r \in (d, \mu)$. Then we have

$$|b_h(P_h w, u_h, \chi_h)| \leq C \|w\|_r \|u_h\|_p \|\nabla \chi_h\|_q \quad (5.8)$$

for $w \in L^r$, $u_h, \chi_h \in X_h$.

Proof. Let $w \in L^r$. Then in view of (4.24) and (4.11), we have that

$$\|G_h P_h w\|_{1,\infty} = \|R_h G w\|_{1,\infty} \leq C \|G w\|_{1,\infty} \leq C \|G w\|_{2,r} \leq C \|w\|_r,$$

which, together with (5.6), yields (5.8). $\square$

LEMMA 5.4 Suppose that (H2) and (R) are satisfied. Let $p \in (d, \mu)$ and $\frac{1}{p} + \frac{1}{q} = 1$. Then,

$$|b(P_h w, u, \chi_h) - b_h(P_h w, \pi_h u, \chi_h)| \leq C h^{1-d/p} \|w\|_p \|u\|_{1,p} \|\nabla \chi_h\|_q \quad (5.9)$$

for $w \in L^p$, $u \in W^{1,p}$ and $\chi_h \in X_h$.

Proof. Let $w \in L^p$, $u \in W^{1,p}$, $\chi_h \in X_h$. We divide

$$\begin{aligned} & b(P_h w, u, \chi_h) - b_h(P_h w, \pi_h u, \chi_h) \\ &= b(P_h w, u, \chi_h - M_h \chi_h) + \{b(P_h w, u, M_h \chi_h) - b_h(P_h w, \pi_h u, \chi_h)\} \\ &\equiv J_1 + J_2. \end{aligned}$$

Because, by (5.2), we have $|J_1| \leq C h \|w\|_p \|u\|_{1,p} \|\nabla \chi_h\|_q$, it is sufficient to prove that

$$|J_2| \leq C h^{1-d/p} \|w\|_p \|u\|_{1,p} \|\nabla \chi_h\|_q. \quad (5.10)$$

To this end, we write as $w_h = P_h w$, $\sigma_{ij}^\pm = \sigma_{ij}^\pm(w_h)$, $u_i = u(P_i)$ and $\chi_i = \chi_h(P_i)$ for the sake of simplicity, and put

$$\hat{\beta}_{ij} = \int_{\Gamma_{ij}} u (\nabla G w_h \cdot v_{ij}) dS, \quad \hat{\beta}'_{ij} = \int_{\Gamma_{ij}} \nabla G w_h \cdot v_{ij} dS.$$

We have

$$b(w_h, u, M_h \chi_h) = \sum_{i=1}^N \chi_i \sum_{j \in \mathcal{A}_i} \int_{\Gamma_{ij}} u (\nabla G w_h \cdot v_{ij}) dS = \sum_{\Gamma_{ij} \in \Gamma_h} (\chi_i - \chi_j) \hat{\beta}_{ij}.$$

This, together with (5.4), implies that

$$\begin{aligned} J_2 &= \sum_{\Gamma_{ij} \in \Gamma_h} (\chi_i - \chi_j) [\hat{\beta}_{ij} - (\sigma_{ij}^+ u_i + \sigma_{ij}^- u_j) \hat{\beta}'_{ij}] \\ &\quad + \sum_{\Gamma_{ij} \in \Gamma_h} (\chi_i - \chi_j) (\sigma_{ij}^+ u_i + \sigma_{ij}^- u_j) (\hat{\beta}'_{ij} - \beta_{ij}(w_h)) \\ &\equiv J_{21} + J_{22}. \end{aligned}$$

To estimate J_{21} , we apply (cf. Lemma 3.1 of Baba & Tabata, 1981)

$$\int_U |v(P) - v(x)| dx \leq Ch_T^{d-d/p} \|v\|_{W^{1,p}(T)} \quad (T \in \mathcal{T}_h, p \in (d, \infty), v \in W^{1,p}),$$

where P is any point in T and U is the intersection of T and any hyper-plane through P . Noting that $\sigma_{ij}^+ + \sigma_{ij}^- = 1$, we have, by (4.3),

$$\begin{aligned} |J_{21}| &\leq \sum_{\Gamma_{ij} \in \Gamma_h} |\chi_i - \chi_j| \cdot \left| \int_{\Gamma_{ij}} [\sigma_{ij}^+(u(x) - u_i) + \sigma_{ij}^-(u(x) - u_j)] (\nabla G w_h \cdot \nu_{ij}) dS \right| \\ &\leq \sum_{\Gamma_{ij} \in \Gamma_h} Ch_{T_{ij}}^{1-d/q} \|\nabla \chi_h\|_{L^q(T_{ij})} \|\nabla G w_h\|_{\infty} h_{T_{ij}}^{d-d/p} \|u\|_{W^{1,p}(T_{ij})} \\ &\leq Ch \|G w_h\|_{2,p} \sum_{T \in \mathcal{T}_h} \|\nabla \chi_h\|_{L^q(T)} \|u\|_{W^{1,p}(T)} \\ &\leq Ch \|P_h w\|_p \|u\|_{1,p} \|\nabla \chi_h\|_q \\ &\leq Ch \|w\|_p \|u\|_{1,p} \|\nabla \chi_h\|_q, \end{aligned}$$

where T_{ij} is defined as (5.7). Now set $v = G P_h w$; then $R_h v = G_h P_h P_h w = G_h P_h w$ by (4.24). Hence, in view of (4.14), we have

$$\|G_h P_h w - G P_h w\|_{1,\infty} \leq Ch^{1-d/p} \|v\|_{2,p} \leq Ch^{1-d/p} \|w\|_p,$$

which implies

$$|\hat{\beta}'_{ij} - \beta_{ij}(w_h)| \leq Ch^{1-d/p} \|w\|_p \text{meas}(\Gamma_{ij}) \leq Ch^{d-d/p} \|w\|_p.$$

Therefore,

$$\begin{aligned} |J_{22}| &\leq \sum_{\Gamma_{ij} \in \Gamma_h} Ch_{T_{ij}}^{1-d/q} \|\nabla \chi_h\|_{L^q(T_{ij})} \|u\|_{\infty} Ch^{d-d/p} \|w\|_p \\ &\leq Ch^{1-d/p} \|u\|_{\infty} \|w\|_p \sum_{T \in \mathcal{T}_h} h^{d/p} \|\nabla \chi_h\|_{L^q(T)} \\ &\leq Ch^{1-d/p} \|u\|_{\infty} \|w\|_p \left(\sum_{T \in \mathcal{T}_h} h^d \right)^{1/p} \|\nabla \chi_h\|_q \\ &\leq Ch^{1-d/p} \|u\|_{1,p} \|w\|_p \|\nabla \chi_h\|_q. \end{aligned}$$

Thus, we obtain (5.10). $\square$

Before concluding this section, we introduce two operators associated with b_h . In view of the second inequality of (5.6) with $p = q = 2$, for a fixed $w_h \in X_h$, we can define the operator $B_{1h}: X_h \rightarrow X_h$ as

$$(v_h, B_{1h} \chi_h) = b_h(w_h, v_h, \chi_h) \quad \forall v_h, \chi_h \in X_h.$$

Then, the operator $B_h: X_h \rightarrow X_h$ is defined as the adjoint operator of B_{1h} in L^2 . Thus,

$$(B_h v_h, \chi_h) = b_h(w_h, v_h, \chi_h) \quad \forall v_h, \chi_h \in X_h. \quad (5.11)$$

Below we will write $B_h(w_h)$ to express the dependence of B_h on w_h .

6. Error estimates (proof of Theorem 2.3)

This section is devoted to the proof of Theorem 2.3. Throughout this section, we suppose that (H1), (H2) and (R) are satisfied, and fix $p \in (d, \mu)$, $\theta \in (1/2, 1)$. Then $q \in (\mu/(\mu - 1), d/(d - 1))$, where $q = p/(p - 1)$. We recall that G is understood as $G = G_p$. We simply write $\alpha_i = \alpha_{i,p}$ for $i = 0, \dots, 3$. Further, the contributions of k and λ are not essential for the discussion below; we take $k = \lambda = 1$ without loss of generality.

Let $u(t)$ be a solution of (2.1), and let $\{u_h^n\}_{n \geq 1}$ be a solution of (2.2), as in Theorem 2.2. First, we shall derive (2.8). We decompose the error into the form of

$$u(t_n) - u_h^n = \eta_h^n + w_h^n \quad (6.1)$$

for $t_n < J$, where $\eta_h^n = \eta_h(t_n)$, $\eta_h(t) = u(t) - U_h(t)$, $U_h(t) = R_h u(t)$, $w_h^n = U_h^n - u_h^n$ and $U_h^n = U_h(t_n)$. Since, from (4.12) and (4.13), we obtain

$$\|\eta_h^n\|_p + h \|\eta_h^n\|_{1,p} \leq C h^2 \alpha_1, \quad (6.2)$$

it suffices to prove that there exist $C_0, h_0, \tau_0 > 0$ such that

$$\|w_h^n\|_p \leq C_0(h^{1-d/p} + \tau^\sigma) \quad (h \in (0, h_0), \tau \in (0, \tau_0)). \quad (6.3)$$

To this end, using (2.2), we observe that

$$\begin{aligned} & (\partial_{\tau_n} w_h^n, \chi_h)_h + (\nabla w_h^n, \nabla \chi_h) + (w_h^n, \chi_h)_h \\ &= (\partial_{\tau_n} U_h^n, \chi_h)_h + (\nabla U_h^n, \nabla \chi_h) + (U_h^n, \chi_h)_h + b_h(u_h^{n-1}, u_h^n, \chi_h) - (u_h^n, \chi_h)_h \end{aligned}$$

for any $\chi_h \in X_h$; equivalently,

$$\begin{aligned} \partial_{\tau_n} w_h^n + A_h w_h^n &= \partial_{\tau_n} U_h^n + A_h U_h^n + K_h^{-1} B_h(u_h^{n-1}) u_h^n - u_h^n \\ &\equiv F^n, \end{aligned} \quad (6.4)$$

where A_h, K_h and $B_h(\cdot)$ are operators defined by (4.22), (4.15) and (5.11), respectively. Thus, putting

$$\begin{aligned} r(\tau_j A_h) &= (I + \tau_j A_h)^{-1} \quad (r(s) = (1 + s)^{-1}), \\ E_{n,j} &= r(\tau_n A_h) r(\tau_{n-1} A_h) \cdots r(\tau_j A_h) \end{aligned}$$

for $1 \leq j \leq n$, we obtain the following identity:

$$w_h^n = E_{n,1} w_h^0 + \sum_{j=1}^n \tau_j E_{n,j} F^j. \quad (6.5)$$

At this stage, we note that (2.1) is reduced to an evolution equation on L^p under (2.5):

$$\frac{du(t)}{dt} + Au(t) + B(u(t))u(t) = u(t) \quad (t \in (0, J)), \quad u(0) = u_0. \quad (6.6)$$

Here, $A = A_p$ is the operator defined by (4.19). For every $w \in L^p$, $B(w) : W^{1,p} \rightarrow L^p$ is defined by $B(w)u = \nabla(u \nabla G w)$. This, together with (4.23), gives

$$\begin{aligned} K_h^{-1} P_h u'(t_j) + A_h U_h^j &= K_h^{-1} P_h u'(t_j) + K_h^{-1} L_h \cdot L_h^{-1} P_h A u(t_j) \\ &= K_h^{-1} P_h [u(t_j) - B(u(t_j))u(t_j)], \end{aligned}$$

where $u' = du/dt$. Hence, F^j is written as

$$\begin{aligned} F^j &= P_h \partial_{\tau_j} (U_h^j - u(t_j)) + P_h (\partial_{\tau_j} u(t_j) - u'(t_j)) + (I - K_h^{-1}) P_h u'(t_j) \\ &\quad + K_h^{-1} P_h [u(t_j) - B(u(t_j))u(t_j)] + K_h^{-1} B_h(u_h^{j-1})u_h^j - u_h^j. \end{aligned}$$

Therefore, we have

$$w_h^n = \sum_{i=0}^4 I_i,$$

where

$$\begin{aligned} I_0 &= E_{n,1} w_h^0 - \sum_{j=1}^n \tau_j E_{n,j} P_h \partial_{\tau_j} \eta_h^j, \\ I_1 &= \sum_{j=1}^n \tau_j E_{n,j} (K_h^{-1} P_h u(t_j) - u_h^j), \\ I_2 &= \sum_{j=1}^n \tau_j E_{n,j} P_h (\partial_{\tau_j} u(t_j) - u'(t_j)), \\ I_3 &= \sum_{j=1}^n \tau_j E_{n,j} (I - K_h^{-1}) P_h u'(t_j), \\ I_4 &= \sum_{j=1}^n \tau_j E_{n,j} K_h^{-1} [B_h(u_h^{j-1})u_h^j - P_h B(u(t_j))u(t_j)]. \end{aligned}$$

First, since

$$I_0 = E_{n,1} P_h (u_0 - u_{0h}) - \sum_{j=1}^{n-1} E_{n,j+1} \tau_j A_h r(\tau_j A_h) P_h \eta_h^j - r(\tau_n A_h) P_h \eta_h^n,$$

we have, by (2.7), (4.8), (4.25), (4.28) and (6.2),

$$\begin{aligned} \|I_0\|_{h,p} &\leq 1 \cdot C \cdot \|u_0 - u_{0h}\|_p + \sum_{j=1}^n 1 \cdot \tau_j \|A_h r(\tau_j A_h)\|_{h,p} C \|\eta_h^j\|_{h,p} + 1 \cdot C \|\eta_h^n\|_{h,p} \\ &\leq \alpha_0 h^{1-d/p} + C(J+1)h^2 \alpha_1. \end{aligned}$$

To estimate I_1 , we put $u^j = u(t_j)$ and $\xi_j = \tau_j(t_n - t_{j-1})^{-\theta}$. We note that

$$\begin{aligned} \|P_h u^j - u_h^j\|_{h,p} &\leq \|P_h u^j - u^j\|_{h,p} + \|u^j - R_h u^j\|_{h,p} + \|w_h^j\|_{h,p} \\ &\leq C h^2 \alpha_1 + \|w_h^j\|_{h,p}; \end{aligned} \tag{6.7}$$

we also note that

$$\sum_{j=1}^n \xi_j \leq \int_0^{t_n} (t_n - s)^{-\theta} ds \leq \frac{J^{1-\theta}}{1-\theta}. \tag{6.8}$$

These, together with (4.27) and (4.34), give

$$\begin{aligned} \|I_1\|_{h,p} &\leq \sum_{j=1}^n \tau_j \|E_{n,j} A_h^\theta\|_{h,p} \|A_h^{-\theta} (K_h^{-1} P_h u^j - u_h^j)\|_{h,p} \\ &\leq C \sum_{j=1}^n \xi_j (\|A_h^{-\theta} (K_h^{-1} - I) P_h u^j\|_{h,p} + \|A_h^{-\theta} (P_h u^j - u_h^j)\|_{h,p}) \\ &\leq C \sum_{j=1}^n \xi_j (h^2 \alpha_1 + \|w_h^j\|_{h,p}) \\ &\leq C \alpha_1 J^{1-\theta} h^2 + C \sum_{j=1}^n \xi_j \|w_h^j\|_{h,p}. \end{aligned}$$

We estimate I_2 as follows:

$$\begin{aligned} \|I_2\|_{h,p} &= \left\| \sum_{j=1}^n E_{n,j} \int_{t_{j-1}}^{t_n} P_h [u'(s) - u'(t_{j-1})] ds \right\|_{h,p} \\ &\leq \sum_{j=1}^n \|E_{n,j}\|_{h,p} \int_{t_{j-1}}^{t_n} C \|u'(s) - u'(t_{j-1})\|_p ds \\ &\leq \sum_{j=1}^n 1 \cdot C \cdot \tau_j^\sigma \alpha_3 \leq C J \tau^\sigma \alpha_3. \end{aligned}$$

Similarly, as in the case of I_1 , we obtain

$$\|I_3\|_{h,p} \leq C J^{1-\theta} h^2 \alpha_2.$$

We proceed to the estimation of I_4 . If it can be shown that

$$\begin{aligned} & \|A_h^{-\theta} K_h^{-1} [B_h(u_h^{j-1})u_h^j - P_h B(u^j)u^j]\|_{h,p} \leq C\alpha_1^2 h^{1-d/p} + C\tau\alpha_1\alpha_2 \\ & + C(1+h^2)\alpha_1\|w_h^{j-1}\|_{h,p} + C\alpha_1\|w_h^j\|_{h,p} + C\|w_h^{j-1}\|_{h,p}\|w_h^j\|_{h,p}, \end{aligned} \quad (6.9)$$

we have

$$\begin{aligned} \|I_4\|_{h,p} & \leq C(\alpha_1^2 + \alpha_2^2)(h^{1-d/p} + \tau)J^{1-\theta} + C\alpha_1 \sum_{j=1}^n \xi_j \|w_h^{j-1}\|_{h,p} \\ & + C(1 + \alpha_1) \sum_{j=1}^n \xi_j (1 + \|w_h^{j-1}\|_{h,p}) \|w_h^j\|_{h,p} \end{aligned} \quad (6.10)$$

in the same manner as for the estimation of I_1 . In order to verify (6.9), it suffices to show that

$$\begin{aligned} |(B_h(u_h^{j-1})u_h^j - B(u^j)u^j, \chi_h)| & \leq [C\alpha_1^2 h^{1-d/p} + C\tau\alpha_1\alpha_2 \\ & + C(1+h^2)\alpha_1\|w_h^{j-1}\|_{h,p} + C\alpha_1\|w_h^j\|_{h,p} \\ & + C\|w_h^{j-1}\|_{h,p}\|w_h^j\|_{h,p}] \|\nabla \chi_h\|_q \end{aligned} \quad (6.11)$$

for $\chi_h \in X_h$. Indeed, by virtue of (4.32),

$$\begin{aligned} & \|A_h^{-\theta} K_h^{-1} [B_h(u_h^{j-1})u_h^j - P_h B(u^j)u^j]\|_{h,p} \\ & \leq C \sup_{\chi_h \in X_h} \frac{(A_h^{-\theta} K_h^{-1} [B_h(u_h^{j-1})u_h^j - P_h B(u^j)u^j], \chi_h)_h}{\|\chi_h\|_{h,q}} \\ & \leq C \sup_{\chi_h \in X_h} \frac{(B_h(u_h^{j-1})u_h^j - B(u^j)u^j, A_h^{-\theta} \chi_h)}{\|\chi_h\|_q}, \end{aligned}$$

which, together with (6.11) and (4.30), implies (6.9). Consequently, (6.9) is reduced to (6.11). To prove (6.11), letting $\chi_h \in X_h$, we write it as

$$\begin{aligned} (B(u^j)u^j - B_h(u_h^{j-1})u_h^j, \chi_h) & = b(u^j, u^j, \chi_h) - b_h(u_h^{j-1}, u_h^j, \chi_h) \\ & = b(u^j, u^j, \chi_h) - b(P_h u^j, u^j, \chi_h) \\ & \quad + b(P_h u^j, u^j, \chi_h) - b_h(P_h u^j, \pi_h u^j, \chi_h) \\ & \quad + b_h(P_h u^j, \pi_h u^j, \chi_h) - b_h(P_h u^{j-1}, \pi_h u^j, \chi_h) \\ & \quad + b_h(P_h u^{j-1}, \pi_h u^j, \chi_h) - b_h(u_h^{j-1}, \pi_h u^j, \chi_h) \\ & \quad + b_h(u_h^{j-1}, \pi_h u^j, \chi_h) - b_h(u_h^{j-1}, u_h^j, \chi_h) \\ & \equiv J_1 + J_2 + J_3 + J_4 + J_5. \end{aligned}$$

We, respectively, obtain, using Lemmas 5.1 and 5.4,

$$|J_1| \leq Ch^2 \alpha_1^2 \|\nabla \chi_h\|_q, \quad |J_2| \leq Ch^{1-d/p} \alpha_1^2 \|\nabla \chi_h\|_q.$$

Furthermore, by virtue of Lemma 5.3 and (6.7), we can deduce the following bounds:

$$\begin{aligned} |J_3| &\leq C \|P_h(u^j - u^{j-1})\|_p \|\pi_h u^j\|_p \|\nabla \chi_h\|_q \\ &\leq C \tau_j \sup_{t \in [0, J]} \|u'(t)\|_p \|u^j\|_\infty \|\nabla \chi_h\|_q \leq C \tau \alpha_1 \alpha_2 \|\nabla \chi_h\|_q, \\ |J_4| &\leq C \|P_h u^{j-1} - u_h^{j-1}\|_p \|\pi_h u^j\|_p \|\nabla \chi_h\|_q \\ &\leq C (h^2 \|u^{j-1}\|_{2,p} + \|w_h^{j-1}\|_{h,p}) \|u^j\|_\infty \|\nabla \chi_h\|_q \\ &\leq C (h^2 \alpha_1 + \|w_h^{j-1}\|_{h,p}) \alpha_1 \|\nabla \chi_h\|_q \end{aligned}$$

and

$$\begin{aligned} |J_5| &\leq C \|u_h^{j-1}\|_p \|\pi_h u^j - u_h^j\|_p \|\nabla \chi_h\|_q \\ &\leq C (\|w_h^{j-1}\|_{h,p} + \|u^{j-1}\|_{1,p}) (h^2 \|u_j\|_{2,p} + \|w_h^j\|_{h,p}) \|\nabla \chi_h\|_q \\ &\leq C (h^2 \alpha_1^2 + h^2 \alpha_1 \|w_h^{j-1}\|_{h,p} + \alpha_1 \|w_h^j\|_{h,p} + \|w_h^{j-1}\|_{h,p} \|w_h^j\|_{h,p}) \|\nabla \chi_h\|_q. \end{aligned}$$

Consequently, we obtain (6.11); hence (6.10) is verified.

Now we can complete the proof of Theorem 2.3. Summing the estimates for $I_0, \dots, I_4$, we obtain

$$\begin{aligned} \|w_h^n\|_{h,p} &\leq C_3 (h^{1-d/p} + \tau^\sigma) + C \alpha_1 \sum_{j=1}^n \zeta_j \|w_h^{j-1}\|_{h,p} \\ &\quad + C(1 + \alpha_1) \sum_{j=1}^n \zeta_j (1 + \|w_h^{j-1}\|_{h,p}) \|w_h^j\|_{h,p}, \end{aligned} \quad (6.12)$$

where $C_3 = C(J+1)(\alpha_1^2 + \alpha_2^2 + \alpha_3)$. We define $z_j = \|w_h^j\|_{h,p}$ and $Z_n = \sup_{0 \leq j \leq n} z_j$. First we assume that

$$Z_{n-1} \leq 1.$$

Thereby, (6.12) implies that

$$z_n \leq C_3 (h^{1-d/p} + \tau^\sigma) + C_4 \sum_{j=1}^n \zeta_j (z_{j-1} + z_j),$$

with $C_4 = C(1 + \alpha_1)$. Hence, by Lemma 4.7,

$$z_n \leq C_5 (h^{1-d/p} + \tau^\sigma) \exp(C_6 J) \equiv \hat{z},$$

where $C_5 = \hat{C} C_3$, $C_6 = \hat{C} C_4^2$ and $\hat{C}$ is the absolute positive constant. Since C_5 and C_6 are independent of h , τ and n , we can choose sufficiently small $h_1 > 0$ and $\tau_0 > 0$ such that $\hat{z} \leq 1$ for $h \in (0, h_1)$ and

$\tau \in (0, \tau_0)$. On the other hand, we choose a sufficiently small $h_2 > 0$ such that $z_0 \leq C\|u_0 - u_{0h}\|_p \leq 1$ for $h \in (0, h_2)$. At this stage, we set $h_0 = \min(h_1, h_2)$. Then, since $h \in (0, h_0)$ and $\tau \in (0, \tau_0)$, we have $Z_n \leq 1$ for all $n \geq 0$ such that $t_n < J$ by induction. In conclusion, we have

$$\|w_h^n\|_p \leq C\|w_h^n\|_{h,p} \leq C[C_5 \exp(C_6 J)](h^{1-d/p} + \tau^\sigma),$$

which implies (6.3), thereby establishing (2.8).

We proceed to the proof of (2.9). We put $v(t) = Gu(t)$, $v_h^n = G_h u_h^n$ and $\hat{w}_h^n = R_h v(t_n) - v_h^n$. Then, by virtue of (2.5) and (4.14),

$$\begin{aligned} \|v(t_n) - v_h^n\|_{1,\infty} &\leq \|v(t_n) - R_h v(t_n)\|_{1,\infty} + \|\hat{w}_h^n\|_{1,\infty} \\ &\leq Ch^{1-d/p} \|u(t_n)\|_p + \|\hat{w}_h^n\|_{1,\infty}. \end{aligned} \quad (6.13)$$

In order to derive an estimate of $\hat{w}_h^n$, we observe that

$$(\nabla \hat{w}_h^n, \nabla \chi_h) + (\hat{w}_h^n, \chi_h) = (P_h(u(t_n) - u_h^n), \chi_h) \quad \forall \chi_h \in X_h,$$

or equivalently,

$$L_h \hat{w}_h^n = P_h(u(t_n) - u_h^n).$$

Hence, in view of (4.31) and (4.8),

$$\|\hat{w}_h^n\|_{1,\infty} \leq C\|L_h \hat{w}_h^n\|_p \leq C\|u(t_n) - u_h^n\|_p. \quad (6.14)$$

Therefore, (2.9) follows from (2.8) and (6.13). This completes the proof of Theorem 2.3. $\square$

7. Remarks on error estimates of the order h

In this section, we suppose that $\Omega \subset \mathbb{R}^d$ is a bounded domain with a sufficiently smooth boundary $\partial\Omega$. Then, (1.1) admits a unique classical solution satisfying

$$u, \frac{\partial u}{\partial x_i}, \frac{\partial^2 u}{\partial x_i \partial x_j}, u_t, v, \frac{\partial v}{\partial x_i}, \frac{\partial^2 v}{\partial x_i \partial x_j} \in C(\overline{\Omega} \times [0, J])$$

and (2.6) holds true for all $p > 0$ and $\sigma = 1$. Moreover, (R) is satisfied for any $\mu > 0$ (see Suzuki & Senba, 2004; Suzuki, 2005).

We take a regular family of (curved) triangulations $\{\mathcal{T}_h\}_h$, which ‘exactly fit the boundary’:

$$\overline{\Omega} = \bigcup_{T \in \mathcal{T}_h} T.$$

The definitions of X_h , $\hat{\phi}_i$, D_i , etc. are similar to those before with obvious modifications (see, e.g. Baba & Tabata, 1981). Under these assumptions, Schatz (1998) proved that

$$\|R_h v - v\|_{1,\infty} \leq Ch\|v\|_{2,\infty} \quad (v \in W^{2,\infty}). \quad (7.1)$$

LEMMA 7.1 Suppose that (H2) is satisfied. Let $p \in (d, \infty)$ with $\frac{1}{p} + \frac{1}{q} = 1$. Then, under the assumptions described above,

$$|b(P_h w, u, \chi_h) - b_h(P_h w, \pi_h u, \chi_h)| \leq Ch(\|w\|_{2,p} + \|Gw\|_{2,\infty})\|u\|_{1,p}\|\nabla \chi_h\|_q \quad (7.2)$$

for $w \in W^{2,p}$, $u \in W^{1,p}$ and $\chi_h \in X_h$.

Proof. Let $w \in W^{2,p}$, $u \in W^{1,p}$, $\chi_h \in X_h$. We follow the notation of the proof of Lemma 5.4. We note first that $Gw \in C^2(\overline{\Omega})$ follows from $w \in C(\overline{\Omega})$. We introduce J_1 , J_{21} and J_{22} , as in the proof of Lemma 5.4. It suffices to derive the estimate for J_{22} . To this end, we decompose it into

$$\begin{aligned} J_{22} &= \sum_{\Gamma_{ij} \in \Gamma_h} (\chi_i - \chi_j)(\sigma_{ij}^+ u_i + \sigma_{ij}^- u_j)(\hat{\beta}'_{ij} - \tilde{\beta}_{ij}) \\ &\quad + \sum_{\Gamma_{ij} \in \Gamma_h} (\chi_i - \chi_j)(\sigma_{ij}^+ u_i + \sigma_{ij}^- u_j)(\tilde{\beta}_{ij} - \beta_{ij}(w_h)) \\ &\equiv J'_{22} + J''_{22}, \end{aligned}$$

where

$$\tilde{\beta}_{ij} = \int_{\Gamma_{ij}} \nabla G w \cdot \nu_{ij} \, dS.$$

Put $v = Gw$; thereby, $R_h v = G_h P_h w$. By virtue of (7.1), we obtain

$$|\tilde{\beta}_{ij} - \beta_{ij}(w_h)| \leq Ch^d \|w\|_{2,\infty};$$

therefore,

$$|J''_{22}| \leq Ch \|Gw\|_{2,\infty} \|u\|_{1,p} \|\nabla \chi_h\|_q.$$

On the other hand,

$$\begin{aligned} |\hat{\beta}'_{ij} - \tilde{\beta}_{ij}| &\leq Ch_{T_{ij}}^{d-1} \|\nabla G(w - P_h w)\|_\infty \\ &\leq Ch_{T_{ij}}^{d-1} \|G(w - P_h w)\|_{2,p} \\ &\leq Ch_{T_{ij}}^{d-1} \|w - P_h w\|_p \\ &\leq Ch_{T_{ij}}^{d-1} h^2 \|w\|_{2,p}. \end{aligned}$$

Hence,

$$|J'_{22}| \leq Ch^2 \|w\|_{2,p} \|u\|_{1,p} \|\nabla \chi_h\|_q.$$

Summing up these estimates, we obtain (7.2). □

With the aid of this lemma, we can derive the following error estimates:

$$\sup_{0 \leq n \leq l} \|u(t_n) - u_h^n\|_p \leq C_1(h + \tau), \quad (7.3)$$

$$\sup_{0 \leq n \leq l} \|Gu(t_n) - G_h u_h^n\|_{1,\infty} \leq C_2(h + \tau) \quad (7.4)$$

for $h \in (0, h_0)$ and $\tau \in (0, \tau_0)$; this can be achieved similarly to the proof of Theorem 2.3. Here, $u(t)$ is a solution of (2.1), $\{u_h^n\}_{n=0}^l \subset X_h$ is a solution of (2.2) as in Theorem 2.2 and $u_{0h} \in X_h$ has been chosen as

$$\|u_0 - u_{0h}\|_p \leq \alpha_{0,p} h$$

with some $\alpha_{0,p} = \alpha_{0,p}(u_0) > 0$. Furthermore, C_i can be taken as

$$C_i = C(J+1) \left(\alpha_{0,p} + \alpha_{1,p}^2 + \alpha_{2,p}^2 + \alpha_{3,p} + \sup_{t \in [0, J]} \|v(t)\|_{2,\infty}^2 \right) \exp[C'(1 + \alpha_{1,p}^2)J],$$

where C, C' are positive constants that depend only on $\Omega, k, \lambda, \gamma_i$'s, h_0 and τ_0 .

However, the exactly fitted triangulation is unrealistic in practical computations. Now we consider approximations of the domain Ω . Let us introduce an approximate polyhedron Ω_h for Ω such that the vertices of $\partial\Omega_h$ are situated on $\partial\Omega$ and assume that there are $h_1 > 0$ and $c_0 > 0$ such that

$$\text{dist}(\Omega, \partial\Omega_h) \leq c_0 h^2 \quad (h \in (0, h_1)). \quad (7.5)$$

In general, $\Omega_h \not\subset \Omega$. Let $\{\mathcal{T}_h\} = \{\mathcal{T}_h\}_{h \downarrow 0}$ be a regular family of triangulations $\mathcal{T}_h$ of Ω_h introduced in Section 2. In this case, $R_h : H^1(\Omega_h) \rightarrow X_h$ is defined by

$$\int_{\Omega_h} \nabla(R_h v - v) \cdot \nabla \chi_h \, dx + \int_{\Omega_h} (R_h v - v) \chi_h \, dx = 0 \quad \forall \chi_h \in X_h.$$

Let $\mathcal{E}$ be a total extension operator for Ω (cf. Adams & Fournier, 2003). That is, for any $m \in \mathbb{N} \cup \{0\}$ and $p \in [1, \infty)$, $\mathcal{E}$ is an operator of $W^{m,p}(\Omega) \rightarrow W^{m,p}(\mathbb{R}^d)$ satisfying

$$(\mathcal{E}u)(x) = u(x) \quad (\text{a.e. } x \in \Omega), \quad \|\mathcal{E}u\|_{W^{m,p}(\mathbb{R}^d)} \leq C_{l,p} \|u\|_{W^{m,p}(\Omega)}.$$

Such a total extension operator extends functions in $C^m(\overline{\Omega})$ to lie in $C^m(\mathbb{R}^d)$. Then, if

$$\|R_h \mathcal{E}v\|_{W^{1,\infty}(\Omega_h)} \leq C \|\mathcal{E}v\|_{W^{1,\infty}(\mathbb{R}^d)} \quad (v \in C^1(\overline{\Omega})), \quad (7.6)$$

$$\|R_h \mathcal{E}v - \mathcal{E}v\|_{W^{1,\infty}(\Omega_h)} \leq Ch \|\mathcal{E}v\|_{W^{2,\infty}(\mathbb{R}^d)} \quad (v \in C^2(\overline{\Omega})), \quad (7.7)$$

we can obtain (7.3) and (7.4) in a similar fashion to that described above. However, unfortunately, it is not obvious that these estimates hold true. They are left here as relevant open problems.

REMARK 7.1 Consider the Dirichlet boundary value problem for the Poisson equation $-\Delta v = f$ in Ω and $v = 0$ on $\partial\Omega$. If Ω is convex, then we actually have (7.6) and (7.7) (see Bakaev *et al.*, 2003).

8. Numerical examples

We assume that $\Omega \subset \mathbb{R}^2$ is a unit square: $\Omega = (0, 1)^2$. We take $\mathcal{T}_h$ as a uniform mesh composed of $2\ell^2$ congruent right-angle triangles for $\ell \in \mathbb{N}$; each side of Ω is divided into ℓ intervals of the same length. Then each small square is decomposed into two equal triangles by a diagonal. We define $h = \sqrt{2}\ell^{-1}$, and we take $k = \lambda = 1$ and $\varepsilon = 0.9$.

Figure 1 shows the results of a numerical experiment at several time levels t_n with the value of $\|u_h^n\|_1$. Therein, the L^1 norm of the initial value u_{0h} is taken as $\|u_{0h}\|_1 = 50 > 8\pi$ so that the corresponding solution u may blow up.

We can observe that a highly concentrated solution is captured successfully. Moreover, the L^1 norm is actually preserved. Next, we plot (t_n, τ_n) in Fig. 2. Comparison of Figs 1 and 2 illustrates that the more the approximate solution is concentrated, the smaller τ_n becomes.

Finally, we consider the rate of convergence. Initial values are taken as $\|u_{0h}\|_1 = 10, 20, 50$, and the shape of u_{0h} is illustrated in Fig. 1(i). (Note that $10, 20 < 8\pi$.) We take $\tau = 10^{-6}$ and $\ell = 8, 16$,

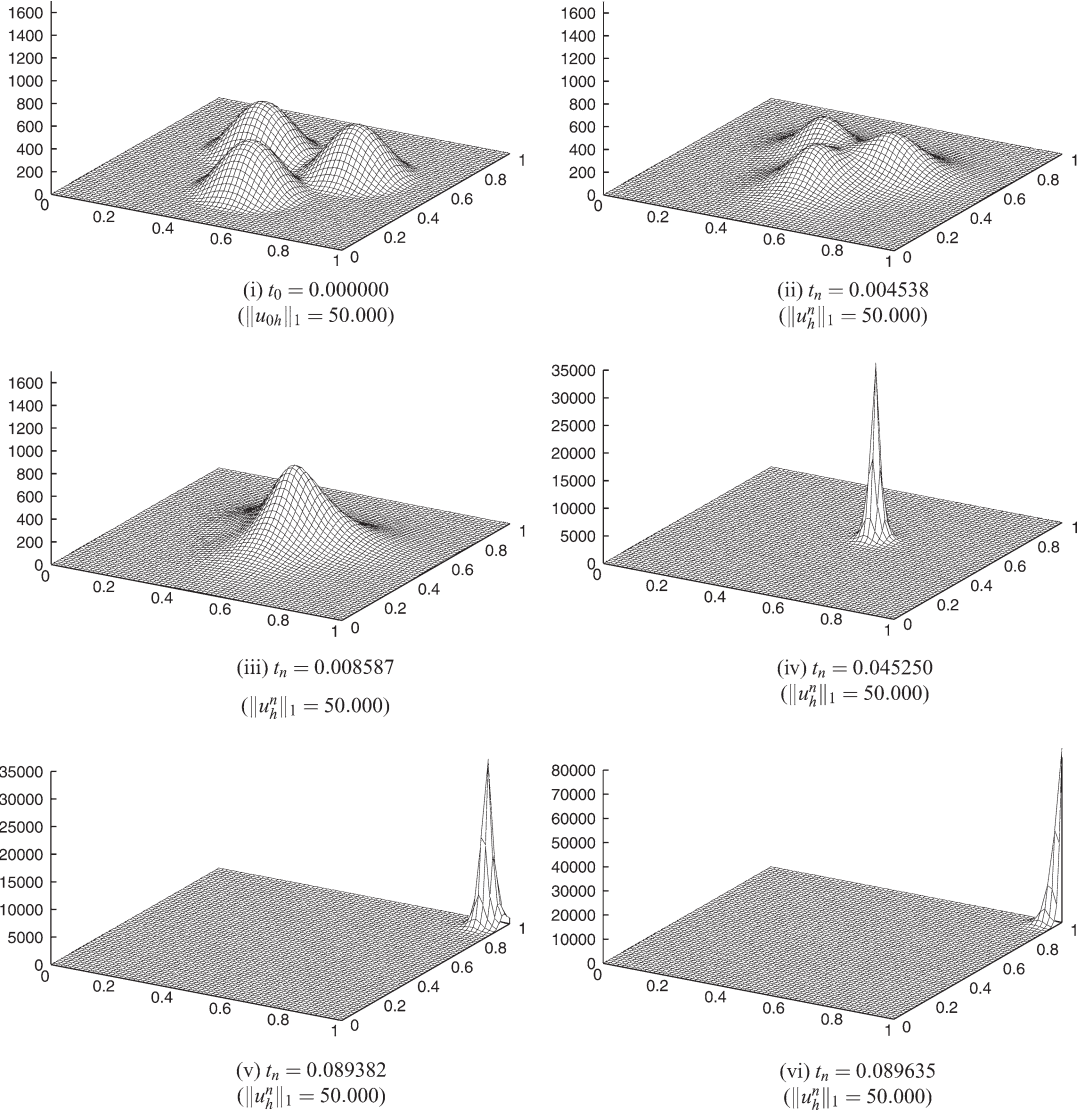


FIG. 1. Shape of u_h^n ($\lambda = k = 1$; $\varepsilon = 0.9$; $\tau = h/2$; $\ell = 64$): (i) the initial function has three peaks; (ii, iii, iv) they gather and produce a single peak; (v, vi) the peak moves towards a corner.

32, 64, 128. Because $\tau/h \ll 1$, we can infer that the effect of time discretization is negligible. The exact solution of (2.1) is not available; therefore, we employ the following technique. We define

$$e_p(h) = \sup_{0 \leq t_n \leq J} \|u_h^n - u_{h/2}^n\|_p$$

with $J = 0.003$. We assume that

$$\sup_{0 \leq t_n \leq J} \|u(t_n) - u_h^n\|_p = Ch^\theta,$$

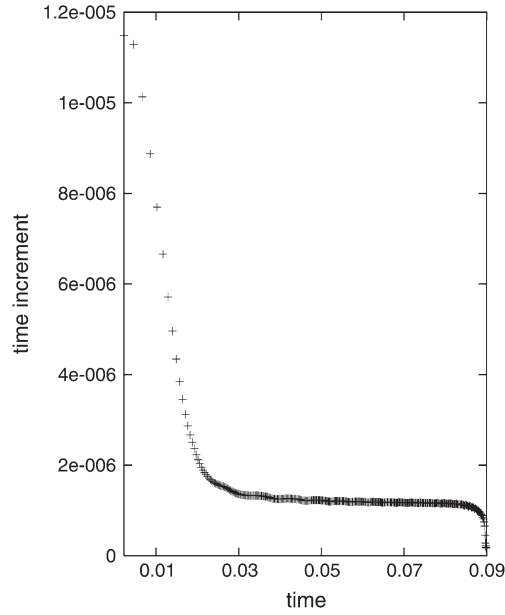


FIG. 2. Time t_n versus time increment τ_n . ($J = 0.09$; $k = \lambda = 1$; $\varepsilon = 0.9$; $\tau = h/2$; $\ell = 64$; $\|u_{0h}\|_1 = 50.0$ and u_{0h} is illustrated in Fig. 1.)

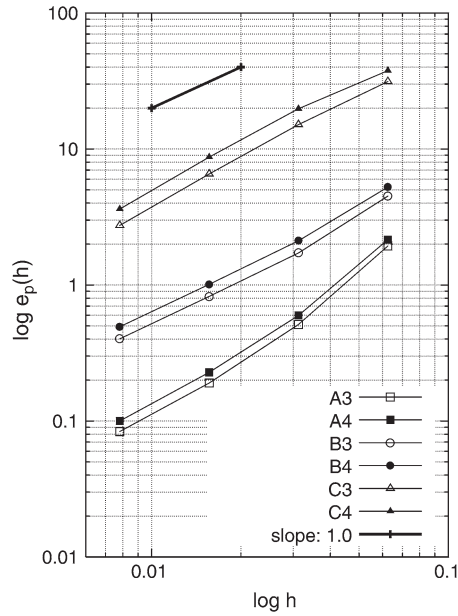


FIG. 3. $\log h$ versus $\log e_p(h)$. ($J = 0.003$; $k = \lambda = 1$; $\varepsilon = 0.9$; $\tau = 10^{-6}$; $\ell = 16, 32, 64, 128$.) A3: $\gamma = 10$, $p = 3$; A4: $\gamma = 10$, $p = 4$; B3: $\gamma = 20$, $p = 3$; B4: $\gamma = 20$, $p = 4$; C3: $\gamma = 50$, $p = 3$; C4: $\gamma = 50$, $p = 4$, where $\gamma = \|u_{0h}\|_1$ and the shape of u_{0h} is illustrated in Fig. 1(i).

where C and θ denote positive constants. Then $e_p(h) \leq C(1 + 2^{-\theta})h^\theta$. In Fig. 3, we plot $(\log h, \log e_p(h))$ for $p = 3, 4$; we see from the figure that $\theta \approx 1$. We note that, in the present situation, (R) with $\mu = \infty$ and (7.1) are satisfied by the Fourier series. Hence, the numerical result supports the argument of Section 6, in particular, the error estimate (7.3). We deduce from this observation that the optimal rate of convergence is actually achieved if u and v are sufficiently regular.

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Appendix

A. Proof of Lemma 4.4

We require some auxiliary results. The first one is a version of the Heinz–Krein inequality on Banach spaces. The second one concerns the dual norm on $W^{1,p}$.

LEMMA A.1 Let A and B , respectively, represent operators of type (ω, M) and (ω', M') on Banach spaces X and Y with bounded inverses (See, e.g. Krein, 1972; Tanabe, 1979). Let P be a bounded linear operator from X into Y . Suppose that $P\mathcal{D}(A) \subset \mathcal{D}(B)$ and that $\|BPu\|_Y \leq C\|Au\|_X$ for all $u \in \mathcal{D}(A)$. Then, for $0 \leq \alpha < \beta \leq 1$, we have $P\mathcal{D}(A^\beta) \subset \mathcal{D}(B^\alpha)$ and $\|B^\alpha Pu\|_Y \leq CK\|A^\beta u\|_X$ for all $u \in \mathcal{D}(A^\beta)$, where $C = C(\alpha, \beta, M, M')$ is positive constant and $K = \max\{1, \|P\|_{X,Y}^{1-\alpha}\}$.

Proof. The proof resembles that of the proof of Lemma 1-7.3 of Krein (1972), where the case of $Y = X$ and $P = I$ was stated explicitly. $\square$

LEMMA A.2 Suppose that (R) is satisfied. Let $p, q \in (1, \mu)$ such that $\frac{1}{p} + \frac{1}{q} = 1$. Then,

$$\|v\|_{1,p} \leq C \sup_{\chi \in W^{1,q}} \frac{|(\nabla v, \nabla \chi) + (v, \chi)|}{\|\chi\|_{1,q}} \quad (\text{A.1})$$

for any $v \in W^{1,p}$.

Proof. Note that (R) guarantees

$$\|u\|_{2,p} \leq C \|A_p u\|_p \quad (u \in \mathcal{D}(A_p)),$$

$$\|u\|_{2,q} \leq C \|A_q u\|_q \quad (u \in \mathcal{D}(A_q)),$$

where A_p and A_q are operators defined as (4.19). Since $A_p^* = A_q$, we can apply Theorem 3.1 of Schechter (1963) ($s = 1, m = 2$) and obtain

$$\|v\|_{1,p} \leq C \sup_{\chi \in W^{1,q}} \frac{|(A_p v, \chi)|}{\|\chi\|_{1,q}} = C \sup_{\chi \in W^{1,q}} \frac{|(\nabla v, \nabla \chi) + (v, \chi)|}{\|\chi\|_{1,q}}$$

for any $v \in \mathcal{D}(A_p)$. Hence, by a density argument, we establish (A.1). $\square$

REMARK A.1 When considering a smooth domain, we can obtain (A.1) for $p \in (1, \infty)$ in a similar manner as in Simader (1972).

Now we can state the following.

Proof of Lemma 4.4. Let $p, q \in (1, \infty)$ such that $\frac{1}{p} + \frac{1}{q} = 1$. We initially note that

$$\|P_h v - v\|_{1,p} \leq Ch \|Av\|_p \quad (v \in \mathcal{D}(A) = \mathcal{W}_p), \quad (\text{A.2})$$

where $A = A_p$. In fact, since $P_h v - R_h v = P_h(v - R_h v)$,

$$\begin{aligned} \|P_h v - v\|_{1,p} &\leq \|P_h v - R_h v\|_{1,p} + \|R_h v - v\|_{1,p} \\ &\leq C \|R_h v - v\|_{1,p} \leq Ch \|v\|_{2,p} \end{aligned}$$

by (4.4) and (4.9), which, together with (2.5), imply (A.2). Let $v \in \mathcal{D}(A)$ and $\chi_h \in X_h$. Then, according to (4.2) and (4.25), we have

$$\begin{aligned} |(A_h P_h v, \chi_h)_h| &= |(\nabla P_h v, \nabla \chi_h) + (P_h v, \chi_h)_h| \\ &\leq |(\nabla(P_h v - v), \nabla \chi_h) + (P_h v - v, \chi_h)| \\ &\quad + |(\nabla v, \nabla \chi_h) + (v, \chi_h)| + |(P_h v, \chi_h)_h - (P_h v, \chi_h)| \\ &\leq \|P_h v - v\|_{1,p} \|\chi_h\|_{1,q} + |(Av, \chi_h)| + Ch^2 \|P_h v\|_{1,p} \|\chi_h\|_{1,q} \\ &\leq Ch \|Av\|_p \cdot Ch^{-1} \|\chi_h\|_q + \|Av\|_p \|\chi_h\|_q + Ch^2 \|Av\|_p \cdot Ch^{-1} \|\chi_h\|_q \\ &\leq C \|Av\|_p \|\chi_h\|_{h,q}; \end{aligned}$$

hence,

$$\|A_h P_h v\|_{h,p} \leq C \|Av\|_p.$$

We can apply Lemma A.1 and obtain, for $0 \leq \alpha < \beta \leq 1$,

$$\|A_h^\alpha P_h v\|_{h,p} \leq C \|A_h^\beta v\|_p \quad (v \in \mathcal{D}(A^\beta)). \quad (\text{A.3})$$

Now let $v_h \in X_h$ and $\theta \in (1/2, 1]$. Then, by (A.3) and (4.20), we have

$$\|A_h^{1-\theta} v_h\|_{h,p} \leq C \|A^{1/2} v_h\|_p \leq C \|v_h\|_{1,p}. \quad (\text{A.4})$$

On the other hand, from (A.1), (4.7) and (4.11), we have

$$\begin{aligned}
 \|v_h\|_{1,p} &\leq C \sup_{\chi \in W^{1,q}} \frac{|(\nabla v_h, \nabla \chi) + (v_h, \chi)|}{\|\chi\|_{1,q}} \\
 &\leq C \sup_{\chi \in W^{1,q}} \frac{|(\nabla v_h, \nabla R_h \chi) + (v_h, R_h \chi)|}{\|R_h \chi\|_{1,q}} \\
 &\leq C \sup_{\chi_h \in X_h} \frac{|(\nabla v_h, \nabla \chi_h) + (v_h, \chi_h)|}{\|\chi_h\|_{1,q}}.
 \end{aligned}$$

Therefore, by virtue of (4.35) and (A.4),

$$\begin{aligned}
 \|v_h\|_{1,p} &\leq C \sup_{\chi_h \in X_h} \left\{ \frac{|(\nabla v_h, \nabla \chi_h) + (v_h, \chi_h)_h|}{\|\chi_h\|_{1,q}} + \frac{|(v_h, \chi_h) - (v_h, \chi_h)_h|}{\|\chi_h\|_{1,q}} \right\} \\
 &\leq C \sup_{\chi_h \in X_h} \left\{ \frac{|(A_h^\theta v_h, \chi_h)_h|}{\|\chi_h\|_{1,q}} + \frac{Ch^2 \|\nabla v_h\|_p \|\nabla \chi_h\|_q}{\|\chi_h\|_{1,q}} \right\} \\
 &\leq C \sup_{\chi_h \in X_h} \frac{|(A_h^\theta v_h, A_h^{1-\theta} \chi_h)_h|}{\|\chi_h\|_{1,q}} + Ch^2 \|v_h\|_{1,p} \\
 &\leq C \|A_h^\theta v_h\|_{h,p} \sup_{\chi_h \in X_h} \frac{\|A_h^{1-\theta} \chi_h\|_{h,q}}{\|\chi_h\|_{1,q}} + Ch^2 \|v_h\|_{1,p} \\
 &\leq C \|A_h^\theta v_h\|_{h,p} + Ch^2 \|v_h\|_{1,p}.
 \end{aligned}$$

Consequently, we have as $h \downarrow 0$,

$$\|v_h\|_{1,p} \leq C \|A_h^\theta v_h\|_{h,p},$$

which completes the proof. $\square$